1. GENERAL

1.1. LOW VISIBILITY PROCEDURES (LVP)

LVP are implemented, when RVR is less than 550m at least at one oft three observation points (touchdown zone, mid-point and stop-end of RWY) and/or ceiling is less than 60m.

RWY 30 is available for take-off in low visibility conditions if LVP are initiated.

1.2. TAXI PROCEDURES

ACFT occupation and taxiing on the maneuvering area shall be carried out by clearance of the appropriate controller TWR or GND.

RWY crossing shall be carried out by TWR controllers clearance only.

Taxiing along TWY A1 on segment from stand 2 to 4 is allowed when stand 1 is vacant.

Taxiing shall be carried out under inboard engines powers.

Taxi routes for towing:

- TWY A2 - RWY 26 - stand 1, when stands 2, 3 and 4 are vacant;
- Strengthened segment in front of RWY 12 THR - RWY 08 - stand 1, when stands 2, 3 and 4 are occupied.

Taxiing of index 4 thru 7 ACFT along RWY 08/26 segment from TWY A2 to intersection line with RWY 12/30 carried out by towing.

Taxiing of index 6 and 7 ACFT under own egines power via MAIN TWY B5 on segment from TWY B9 to TWY B5A is prohibited.

1.3. PARKING INFORMATION

Stands 16 thru 17BR, 18A thru 18BR, 19A thru 19BR and 20A thru 20BR are available for helicopters.

ACFT taxiing or towing to/from stands 16 thru 20B shall be carried out after Follow-me car.

Taxiing of ACFT out of stands shall be carried out by towing.

1.4. NOISE ABATEMANT PROCEDURES

Noise abatement procedures shall not be executed at the expense of reduction of flight safety.

Deviation from the procedures can be permitted only by the reason of flight safety.

A displaced RWY THR shell not be used as a noise abatement measure.

1.4.1. USE OF RWY SYSTEM DURING THE NIGHT PERIOD

RWY 30 is available for take-off and landing of Tu-134, Tu-154, Il-86, Il-76, An-12, An-26 ACFT types not complying with noise level requirements of ICAO Annex 16, except Head of State flights and official delegations flights, ambulance flights, search and rescue flights.

1.5. OTHER INFORMATION

Birds in vicinity of APT.

Flight crews are recommended to swich on landing lights to scare off the birds.

2. ARRIVAL

2.1. COMMUNICATION FAILURE PROCEDURES

2.1.1. COMMUNICATION FAILURE DURING ARRIVAL

Continue the flight maintaining flight route and profile of the cleared (shortest basic) RNAV STAR to the maximum extent.

Execute approach in accordance with the established procedure.

In case of missed approach carry out published missed approach procedure to the maximum extent to the nearest holding area, and follow paragraph 2.1.2.

If deviation from the specified procedure is required flight crew must set transponder to code 7700.

2.1.2. COMMUNICATION FAILURE DURING/AFTER MISSED APPROACH

Proceed to the nearest holding area, maintaining flight route and profile of the missed approach procedure to the maximum extent.

Enter the holding area at the upper published altitude at IAF, burn out fuel, if necessary.

After taking the decision to execute landing at the destination aerodrome:

- Execute approach in accordance with the established procedure.

After taking the decision to proceed to an alternate aerodrome in Moscow TMA:

- Proceed to RT VORDME climbing to transition altitude 10000'.
- Proceed to IAF of the alternate aerodrome in Moscow TMA via the following waypoints:

  Moscow/Sheremetyevo:

  ANDIF - GEKLA - RUGEL - BESTA - SORET - RIMDE - NDB KN - EE043 - EE044 - AGMER - EE045 - TAFAZ - KEZVU (IAF).

  Moscow/Domodedovo:

  ANDIF - IMZUP - KUPVE - NIDBE - IZVOK - IPKED - ZOVGO - ODZAG - GUFUZ - ALBOR (IAF).

  Moscow/Vnukovo:

  ANDIF - IMZUP - GEGNA - KIBUR - NDB LO - BEMAS - TEBDI - TEPTA - RONEZ - TOLKE - TADUT - FIDOT - RORUK (IAF).

  Ostafyevo:

  ANDIF - IMZUP - GEGNA - KIBUR - NDB LO - BEMAS - TEBDI - TEPTA - RONEZ - TOLKE - TADUT - FIDOT - RORUK (IAF).
- At IAF enter the published, if available, or standard holding area.
- In the holding area descend from transition altitude 10000' to the upper published approach procedure altitude at IAF.
- Execute approach in accordance with the established procedure.

After taking the decision to proceed to an alternate aerodrome indicated in the flight plan outside Moscow TMA:

- Execute approach in accordance with the established procedure to IF.
- Proceed from IF to the initiation point of the basic RNAV SID of the same RWY.
- Maintain flight route and profile of the basic RNAV SID to the maximum extent until leaving Moscow TMA.
- After leaving Moscow TMA reach the flight level specially established for flight without radio communication (FL140, FL150, FL240, FL250).

2. ARRIVAL

2.2. SPEED RESTRICTIONS

280 KT ±10 KT or Mach 0.8 whichever is less from cruising FL to FL 250;
270 KT ±10 KT below FL 250 to FL 100;
250 KT ±10 KT below FL 100 to TL.

Below transition level - IAS shall be as in accordance with the Aeroplane Flight Manual without exceeding IAS indicated in limitations section. The above mentioned IAS shall be maintained by the flight crew unless otherwise instructed by ATS unit or by the established approach procedure.

If unable to maintain the above mentioned IAS, the flight crew shall report it to ATS unit.

2.3. NOISE ABATEMENT PROCEDURES

2.3.1. NOISE ABATEMENT APPROACH PROCEDURES

Operational noise abatement procedures on the stage of approach shall be carried out by the crews of all ACFT.

RWY 30 is a noise preferential RWY which is maximally used for ACFT landings when possible.

The flight below ILS GP slope is PROHIBITED during instrument and visual approach.

Straight-in Approach RWY 30

The flight crew shall carry out the flight at 3350'/900m by the moment when the ACFT reaches the distance of 13.5NM to RWY THR, maintaining indicated airspeed of 210 KT and the flight direction which allows entering the coverage area of the ILS LOC providing approach.

At a distance of 11.9NM to RWY THR, the flight crew shall reduce indicated airspeed to 185 KT (±10 KT) and carry out descent to 2000'/500m to enter the ILS LOC coverage area at a distance of 8.1NM to RWY THR at 2000'/500m.

At a distance of 8.1NM to RWY THR, the flight crew shall extend landing gear.

The extension of wing high-lift devices into the intermediate position shall be carried out before intercepting the GP.

The flight crew shall continue the reduction of indicated airspeed after intercepting the GP at 2000'/500m and initiation of descent along the GP to establish the indicated airspeed of extension of wing high-lift devices into landing configuration by the moment when the ACFT reaches 1700'/400m.

During descent along the GP at 1700' /400m the flight crew shall continue extension of wing high-lift devices into the landing configuration and establish the speed of the final approach.

After that the speed shall be maintained according to the Aeroplane Flight Manual.

3. DEPARTURE

3.1. NOISE ABATEMENT PROCEDURES

3.1.1. TAKE-OFF AND CLIMBING PROCEDURE

General

RWY 12 is a noise preferential RWY.

Noise abatement procedures shall not be executed in case of one of the ACFT engines failure during take-off phase.

The initial climb speed to the noise abatement initiation point shall not be less than V_2+10 KT/20km/h.

ACFT power reduction shall not be applied until:

- The ACFT reaches 1200'/240m.
- The set standard power mode allows to maintain the steady climb gradient not less than 4.0% at the speed indicated above with maximum certified take-off mass.

Take-off path provides passing over all obstacles located under the flight path with sufficient clearance with all engines operative as well as taking into account the possibility of engine failure and the period of time required for the development of full power by the remaining engines.

Restrictions

- Take-off shall be executed at the take-off mode of engines operation.
- High lift devices shall be set into take-off configuration.

The noise abatement procedure shall be initiated at not less than 1200'/240m.

On reaching 1200'/240m adjust and maintain engines power/thrust in accordance with the noise abatement power/thrust adjustment schedule provided in the Aeroplane Flight Manual.

Maintain a climb speed of V_2 + (10-20 KT)/(20-40km/h) with flaps in the take-off configuration.

At 3350'/900m, while maintaining a positive rate of climb, accelerate and retract flaps on schedule: indicated airspeed - angle of flaps and slats deflection (taking into account ACFT mass).

Above 3350'/900 m execute smooth acceleration to en-route climb speed.

3.2. COMMUNICATION FAILURE PROCEDURES

3.2.1. COMMUNICATION FAILURE DURING DEPARTURE

Continue the flight maintaining flight route and profile of the cleared RNAV SID to the maximum extent.

After taking the decision to return to the aerodrome the departure was taken from:

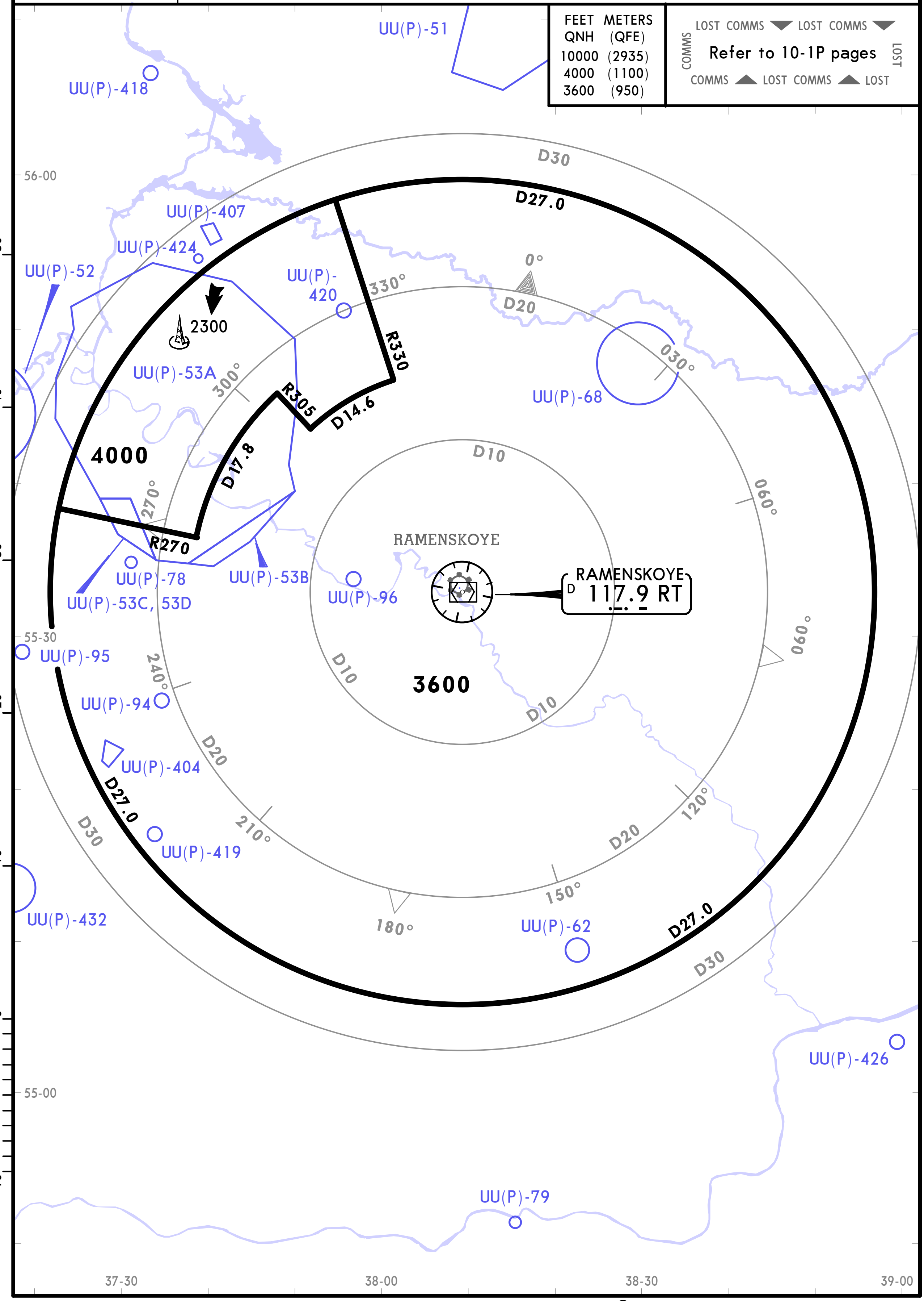
- Proceed to SID termination fix and then to the nearest origination point of the shortest basic RNAV STAR of the departure aerodrome.
- Maintain flight route and profile of the basic RNAV STAR to the maximum extent.
- Execute approach in accordance with the established procedure.
- In case of a missed approach proceed to the nearest holding area, maintaining flight route and profile of the missed approach procedure to the maximum extent and follow paragraph 2.1.2.

After taking the decision to proceed to the destination aerodrome:

- After leaving Moscow TMA continue climbing to the flight level indicated in the flight plan.
- If deviation from the specified procedure is required, set transponder to code 7700.

RADAR MINIMUM ALTITUDES

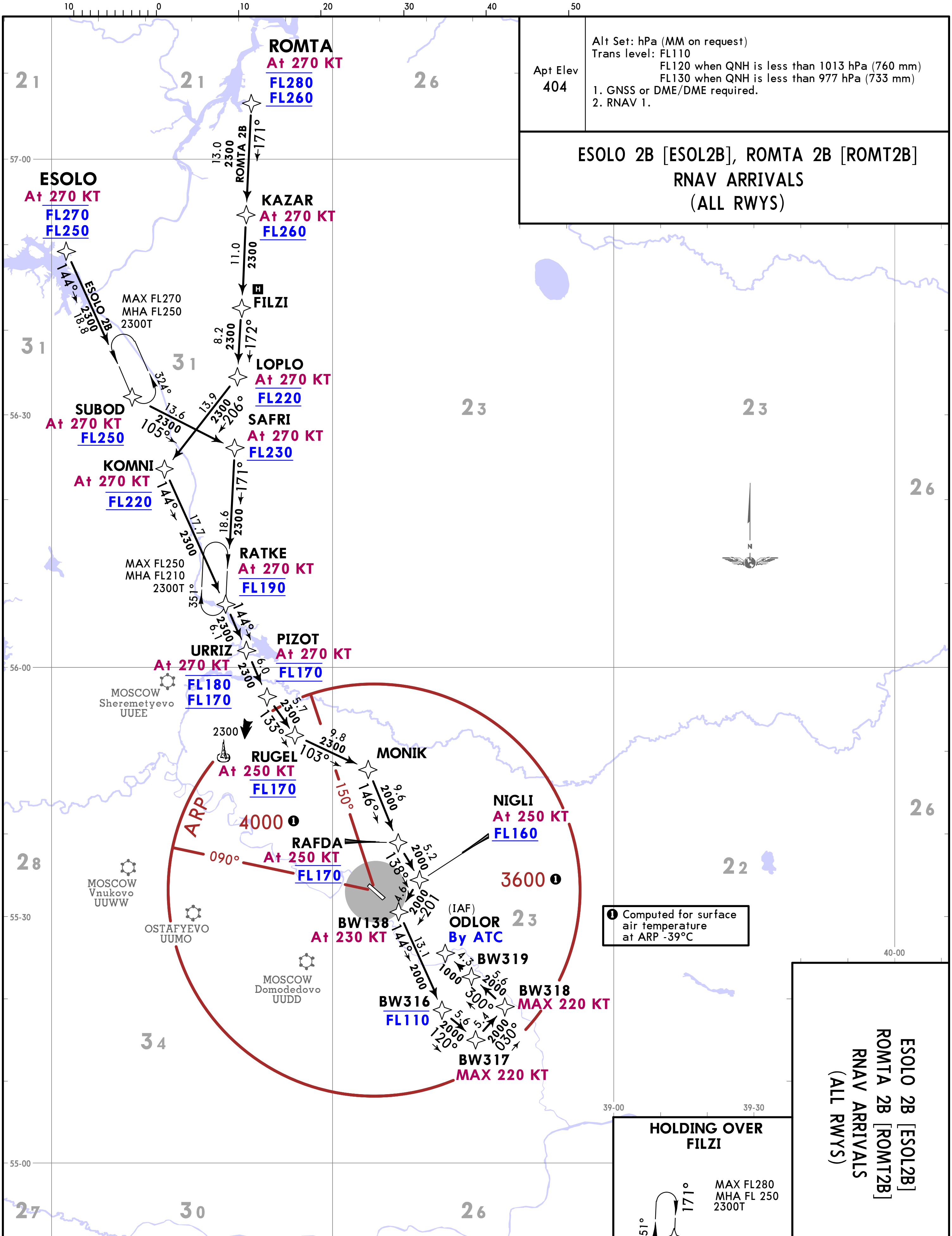
GORDY Radar (TWR) 125.25	Alt Set: hPa (MM on request) Trans level: FL110 FL120 when QNH is less than 1013 hPa (760 mm) FL130 when QNH is less than 977 hPa (733 mm) Trans alt: 10000 QNH (QFE on request)
Apt Elev 404	1. Chart only to be used for cross-checking of altitudes assigned while under vectoring control. 2. Flight levels assigned by ATC take into account corrections for low temperature effect.



CHANGES: Sector altitudes revised, prohibited areas established.

Apt Elev 404	Alt Set: hPa (MM on request) Trans level: FL110 FL120 when QNH is less than 1013 hPa (760 mm) FL130 when QNH is less than 977 hPa (733 mm) 1. GNSS or DME/DME required. 2. RNAV 1.

ESOLO 2B [ESOL2B], ROMTA 2B [ROMT2B] RNAV ARRIVALS (ALL RWYS)

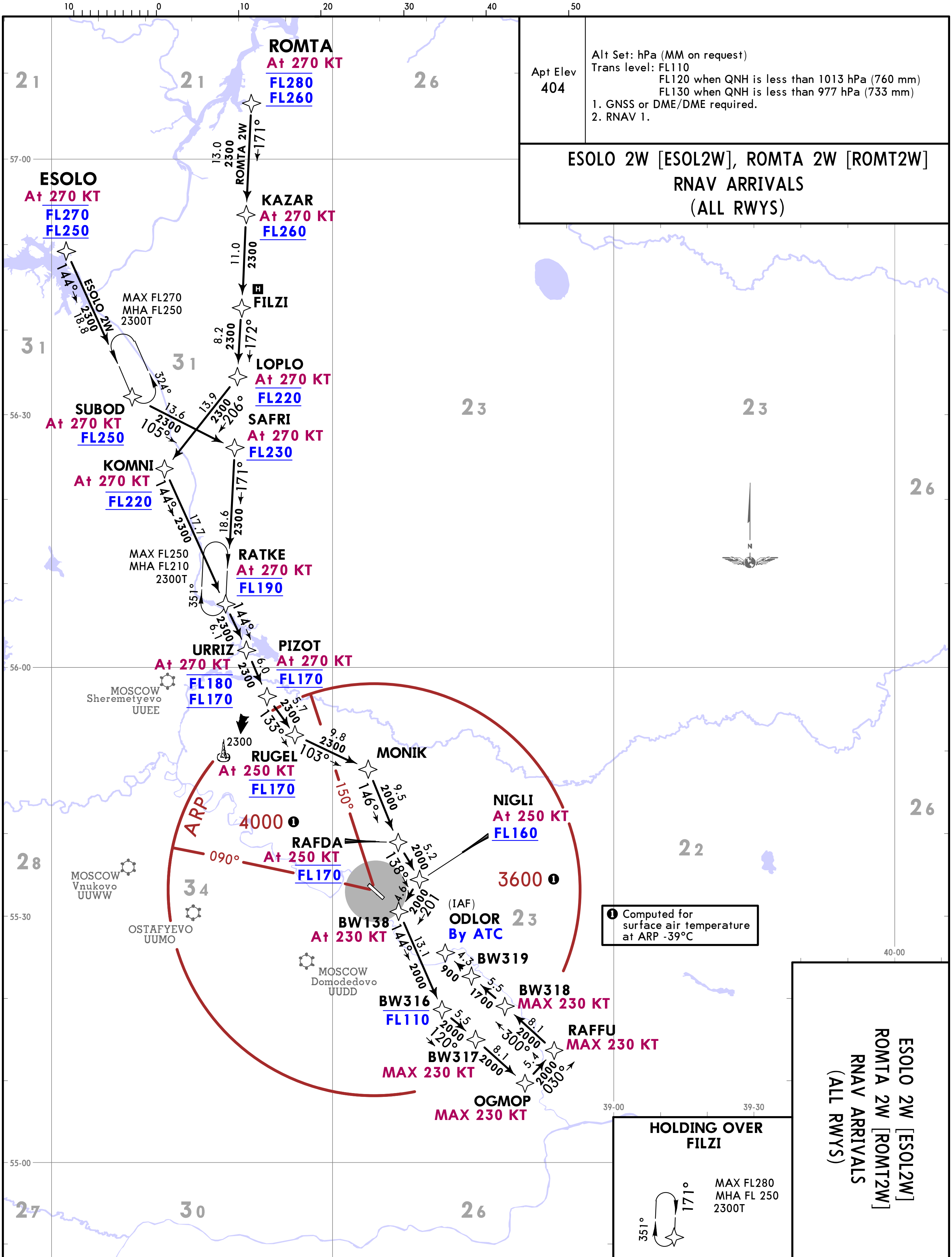


ESOLO 2B [ESOL2B]
ROMTA 2B [ROMT2B]
RNAV ARRIVALS
(ALL RWYS)

CHANGES: MSA raised.

Apt Elev	
404	Alt Set: hPa (MM on request) Trans level: FL110 FL120 when QNH is less than 1013 hPa (760 mm) FL130 when QNH is less than 977 hPa (733 mm) 1. GNSS or DME/DME required. 2. RNAV 1.

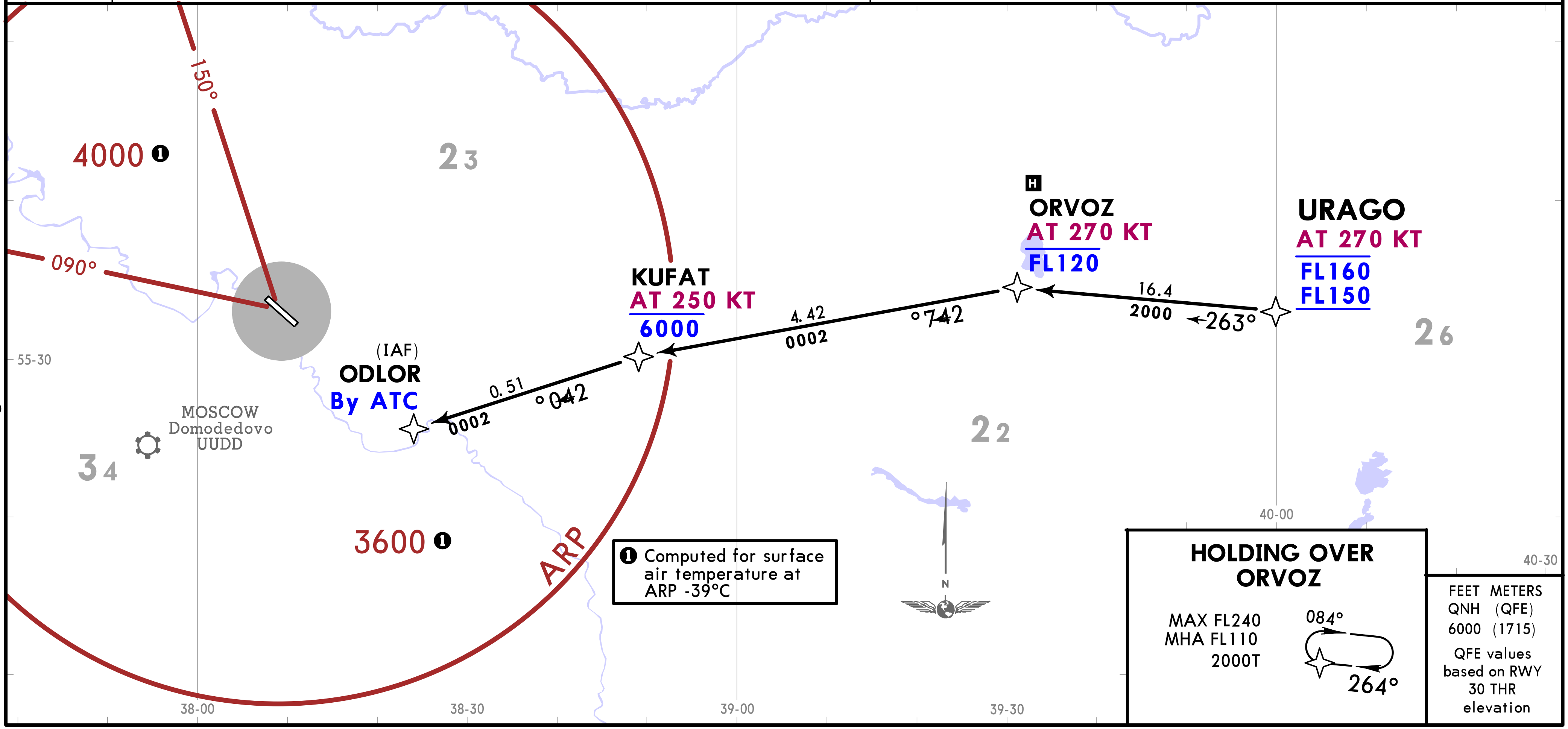
ESOLO 2W [ESOL2W], ROMTA 2W [ROMT2W] RNAV ARRIVALS (ALL RWYS)



URAGO 2B [URAG2B]
RNAV ARRIVAL
(ALL RWYS)

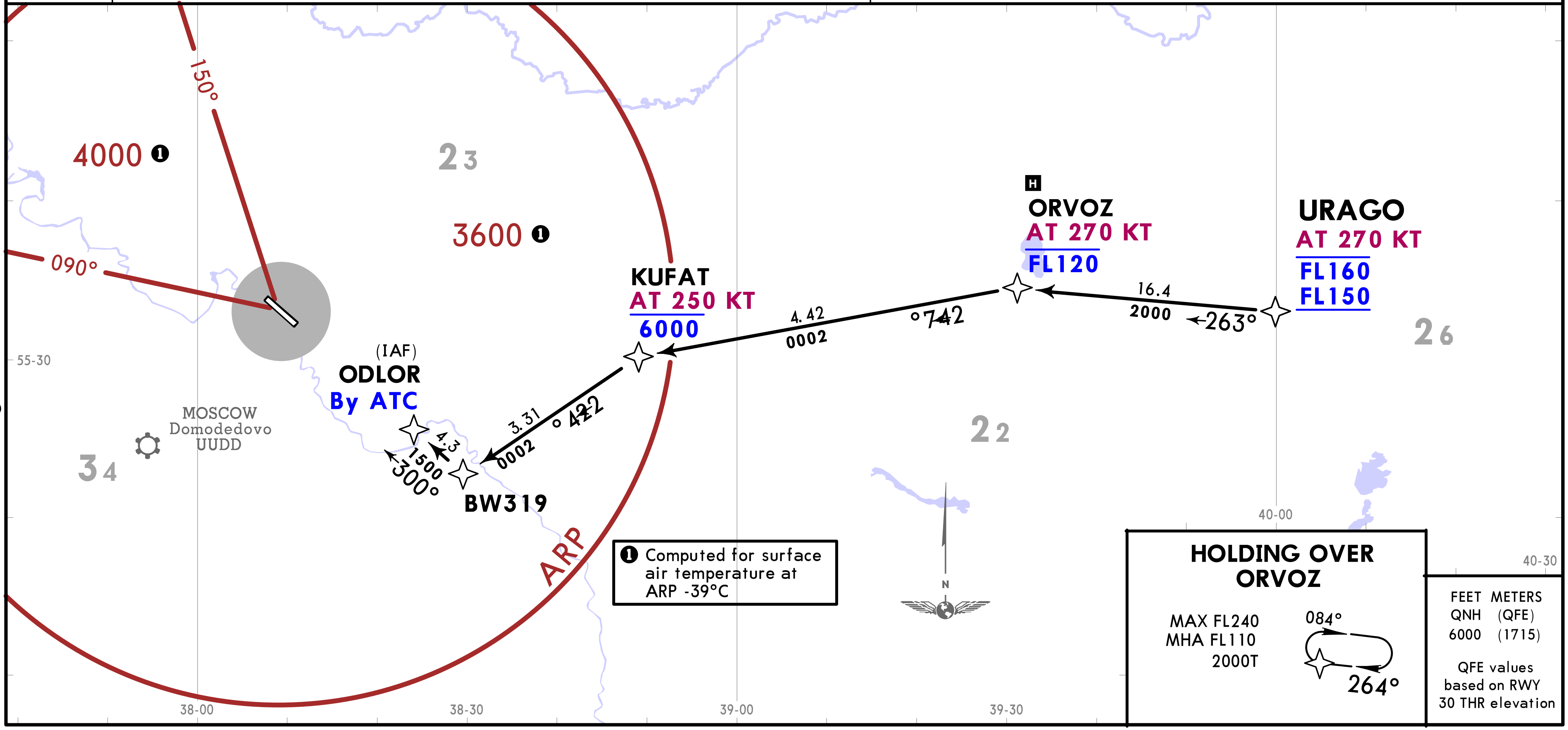
Apt Elev
404

Alt Set: hPa (MM on request)
Trans level: FL110
FL120 when QNH is less than 1013 hPa (760 mm)
FL130 when QNH is less than 977 hPa (733 mm)
1. GNSS or DME/DME required.
2. RNAV 1.



CHANGES: MSA raised.

Apt Elev 404	Alt Set: hPa (MM on request) Trans level: FL110 FL120 when QNH is less than 1013 hPa (760 mm) FL130 when QNH is less than 977 hPa (733 mm) 1. GNSS or DME/DME required. 2. RNAV 1.	URAGO 2W [URAG2W] RNAV ARRIVAL (ALL RWYS)

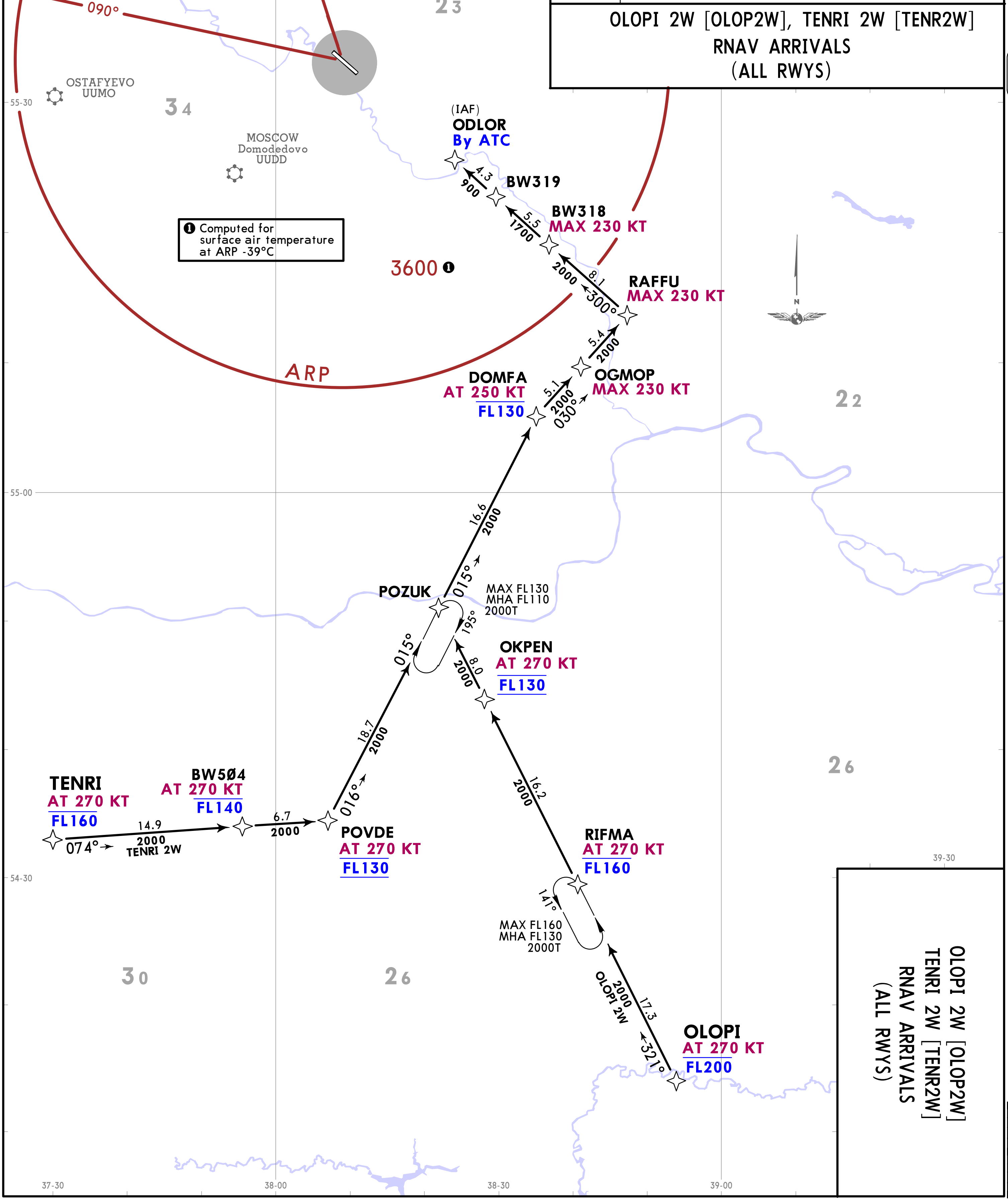


CHANGES: MSA raised.

CHANGES: MSA raised.

Apt Elev 404	Alt Set: hPa (MM on request) Trans level: FL110 FL120 when QNH is less than 1013 hPa (760 mm) FL130 when QNH is less than 977 hPa (733 mm) 1. GNSS or DME/DME required. 2. RNAV 1.

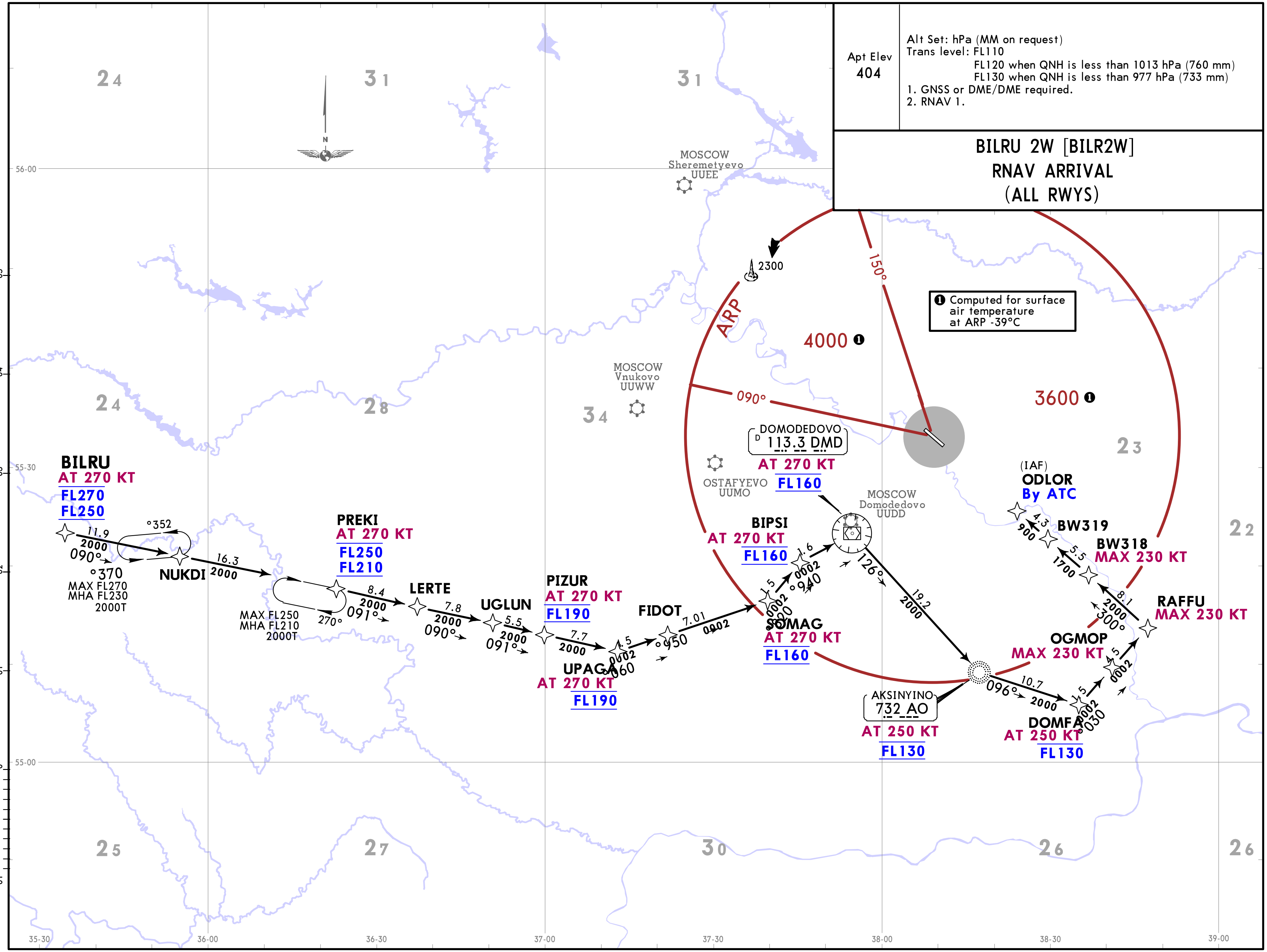
OLOPI 2W [OLOP2W], TENRI 2W [TENR2W]
RNAV ARRIVALS
(ALL RWYS)



OLOPI 2W [OLOP2W]
TENRI 2W [TENR2W]
RNAV ARRIVALS
(ALL RWYS)

Apt Elev	
404	Alt Set: hPa (MM on request) Trans level: FL110 FL120 when QNH is less than 1013 hPa (760 mm) FL130 when QNH is less than 977 hPa (733 mm) 1. GNSS or DME/DME required. 2. RNAV 1.

BILRU 2W [BILR2W]
RNAV ARRIVAL
(ALL RWYS)



CHANGES: MSA raised.

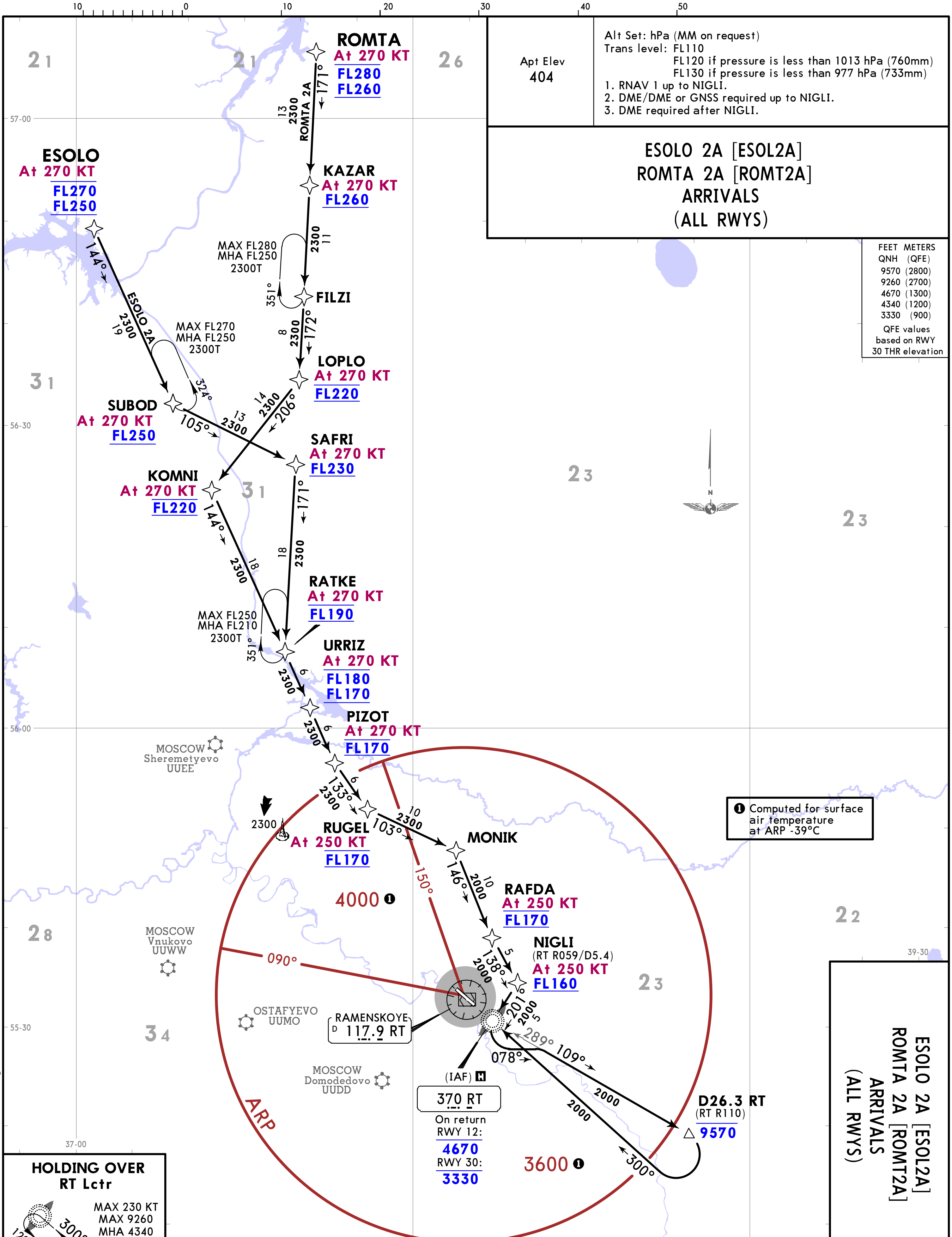
CHANGES: MSA raised.

Apt Elev 404	Alt Set: hPa (MM on request) Trans level: FL110 FL120 if pressure is less than 1013 hPa (760mm) FL130 if pressure is less than 977 hPa (733mm) 1. RNAV 1 up to NIGLI. 2. DME/DME or GNSS required up to NIGLI. 3. DME required after NIGLI.

ESOLO 2A [ESOL2A]
ROMTA 2A [ROMT2A]
ARRIVALS
(ALL RWYS)

FEET QNH	METERS (QFE)
9570	(2800)
9260	(2700)
4670	(1300)
4340	(1200)
3330	(900)

QFE values based on RWY 30 THR elevation

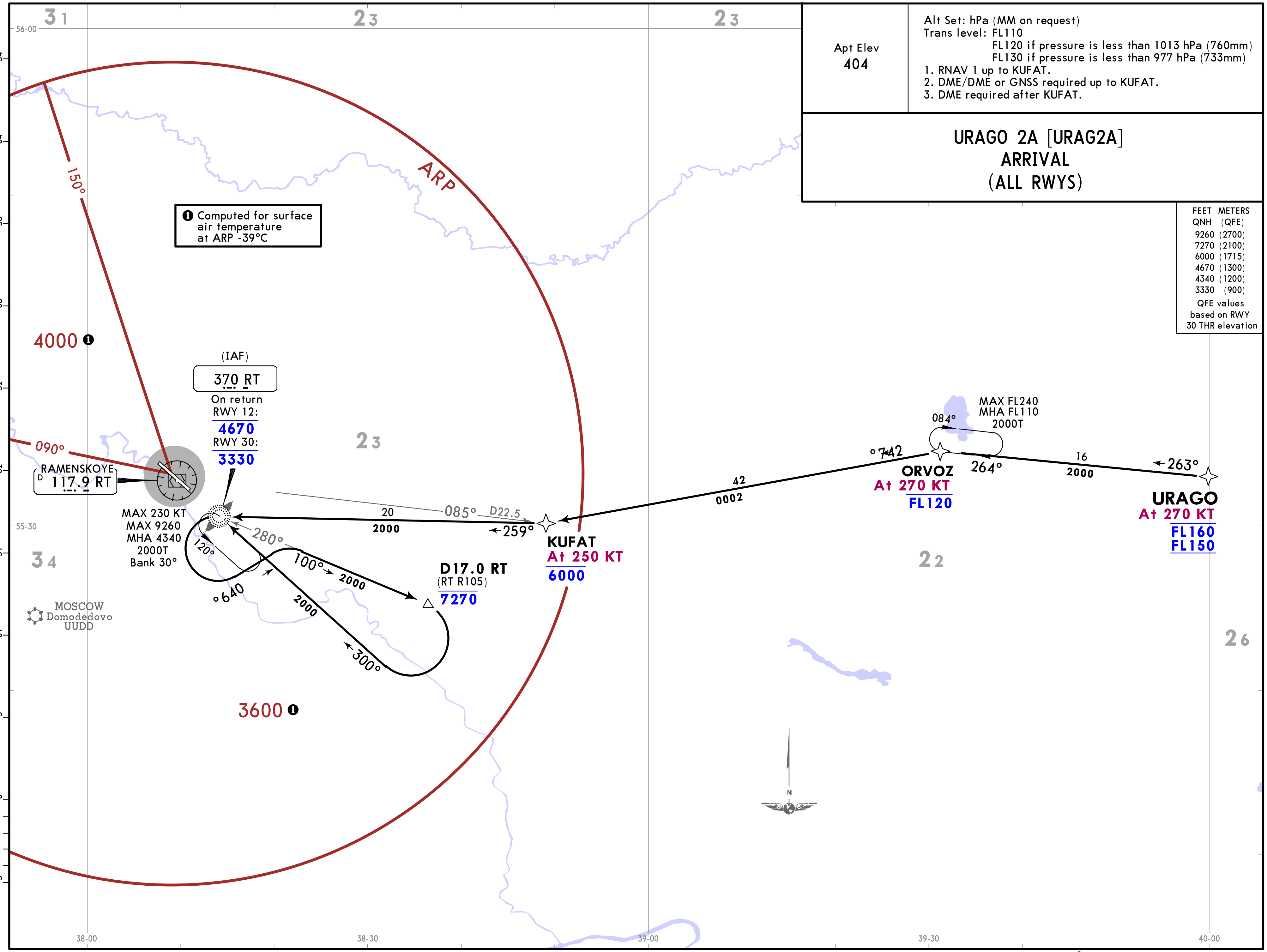


Apt Elev 404	Alt Set: hPa (MM on request) Trans level: FL110 FL120 if pressure is less than 1013 hPa (760mm) FL130 if pressure is less than 977 hPa (733mm) 1. RNAV 1 up to KUFAT. 2. DME/DME or GNSS required up to KUFAT. 3. DME required after KUFAT.

URAGO 2A [URAG2A] ARRIVAL (ALL RWYS)

FEET QNH	METERS (QFE)
9260	(2700)
7270	(2100)
6000	(1715)
4670	(1300)
4340	(1200)
3330	(900)

QFE values based on RWY 30 THR elevation



CHANGES: MSA raised.

STAR

CHANGES: MSA raised.

Apt Elev	
404	Alt Set: hPa (MM on request) Trans level: FL110 FL120 if pressure is less than 1013 hPa (760mm) FL130 if pressure is less than 977 hPa (733mm) 1. RNAV 1 up to DOMFA. 2. DME/DME or GNSS required up to DOMFA. 3. DME required after DOMFA.

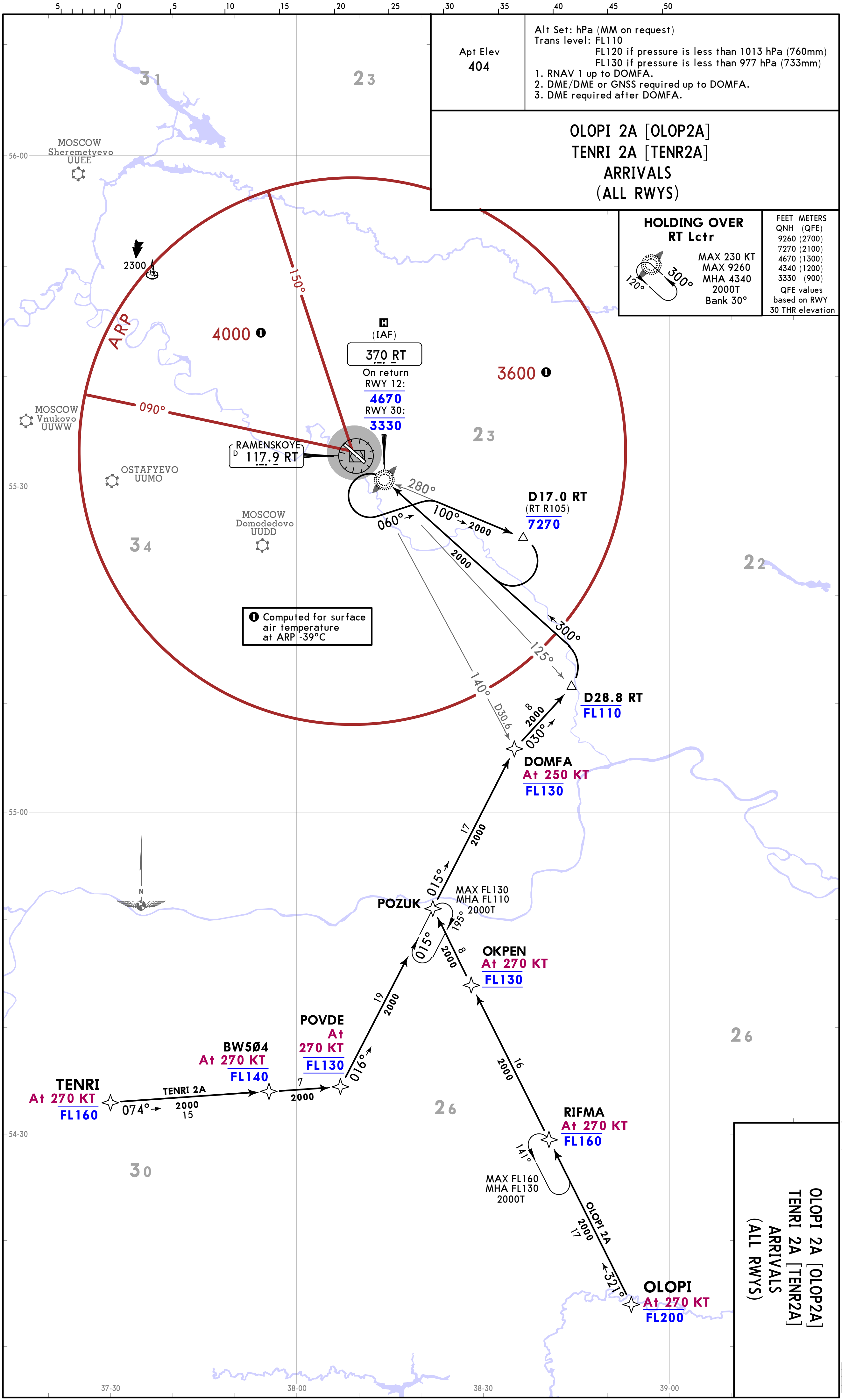
OLOPI 2A [OLOP2A]
TENRI 2A [TENR2A]
ARRIVALS
(ALL RWYS)

HOLDING OVER RT Lctr

MAX 230 KT
MAX 9260
MHA 4340
2000T
Bank 30°

FEET QNH	METERS (QFE)
9260	(2700)
7270	(2100)
4670	(1300)
4340	(1200)
3330	(900)

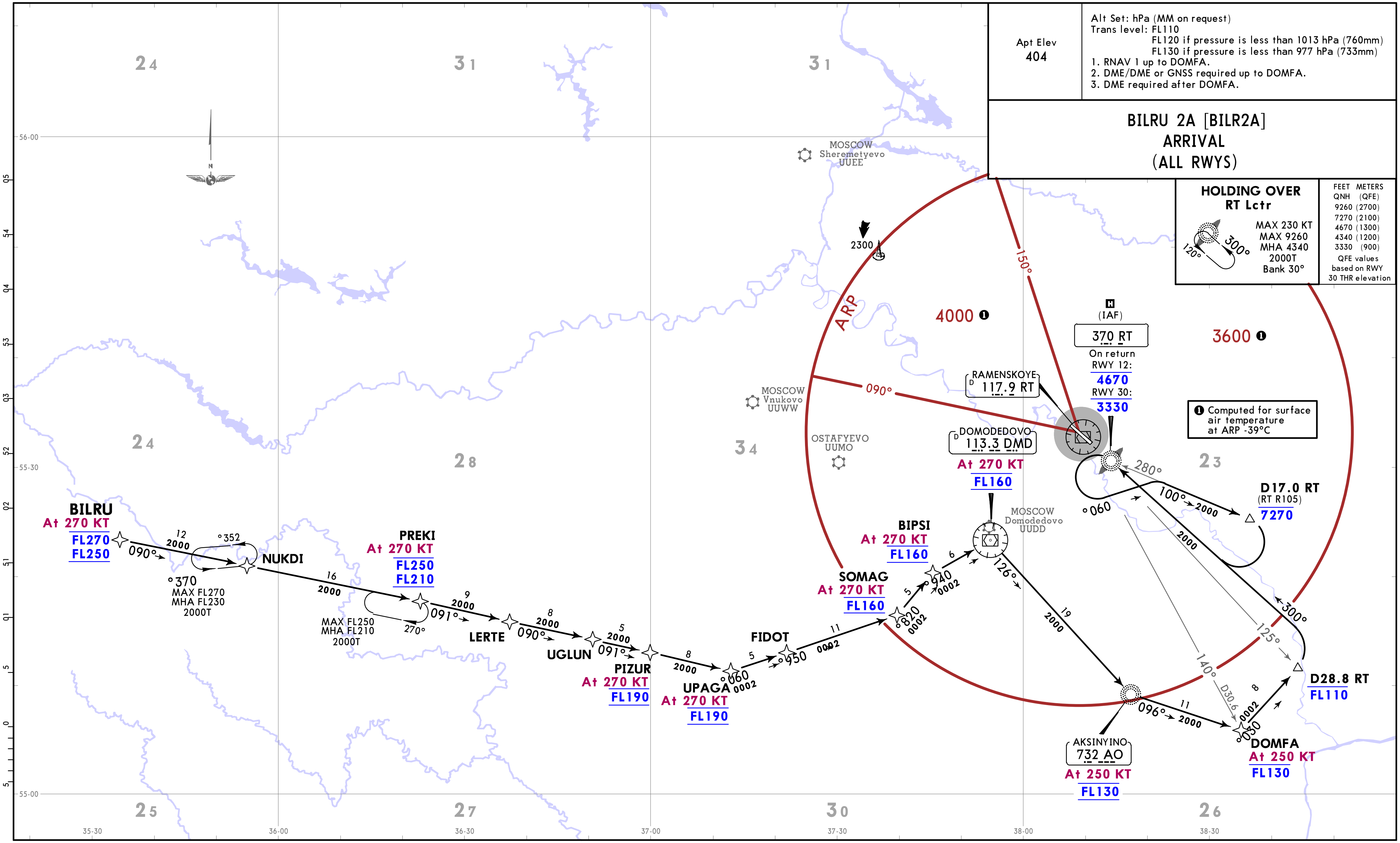
QFE values based on RWY 30 THR elevation



OLOPI 2A [OLOP2A]
TENRI 2A [TENR2A]
ARRIVALS
(ALL RWYS)

Apt Elev 404	Alt Set: hPa (MM on request) Trans level: FL110 FL120 if pressure is less than 1013 hPa (760mm) FL130 if pressure is less than 977 hPa (733mm) 1. RNAV 1 up to DOMFA. 2. DME/DME or GNSS required up to DOMFA. 3. DME required after DOMFA.

BILRU 2A [BILR2A]
ARRIVAL
(ALL RWYS)



CHANGES: MSA raised.

UUBW/ZIA RAMENSKOYE

Apt Elev 404

Trans alt: 10000 QNH (QFE on request)

1. RNAV 1.
2. GNSS or DME/DME required.
3. Turn before DER is prohibited.
4. If unable to comply with published restrictions, advise ATC.

KOGOM 2B [KOGO2B]
KOGOM 2W [KOGO2W]
OLMUN 2B [OLMU2B]
OLMUN 2W [OLMU2W]
RILPO 2B [RILP2B]
RILPO 2W [RILP2W]
RNAV DEPARTURES (ALL RWYS)

Close-in obstacles

MAX elev 448 between 0.1 and 0.2 NM from RWY 12 DER located to the left and to the right of take-off heading.
MAX elev 491 between 0.3 and 0.5 NM from RWY 30 DER located to the left and to the right of take-off heading.

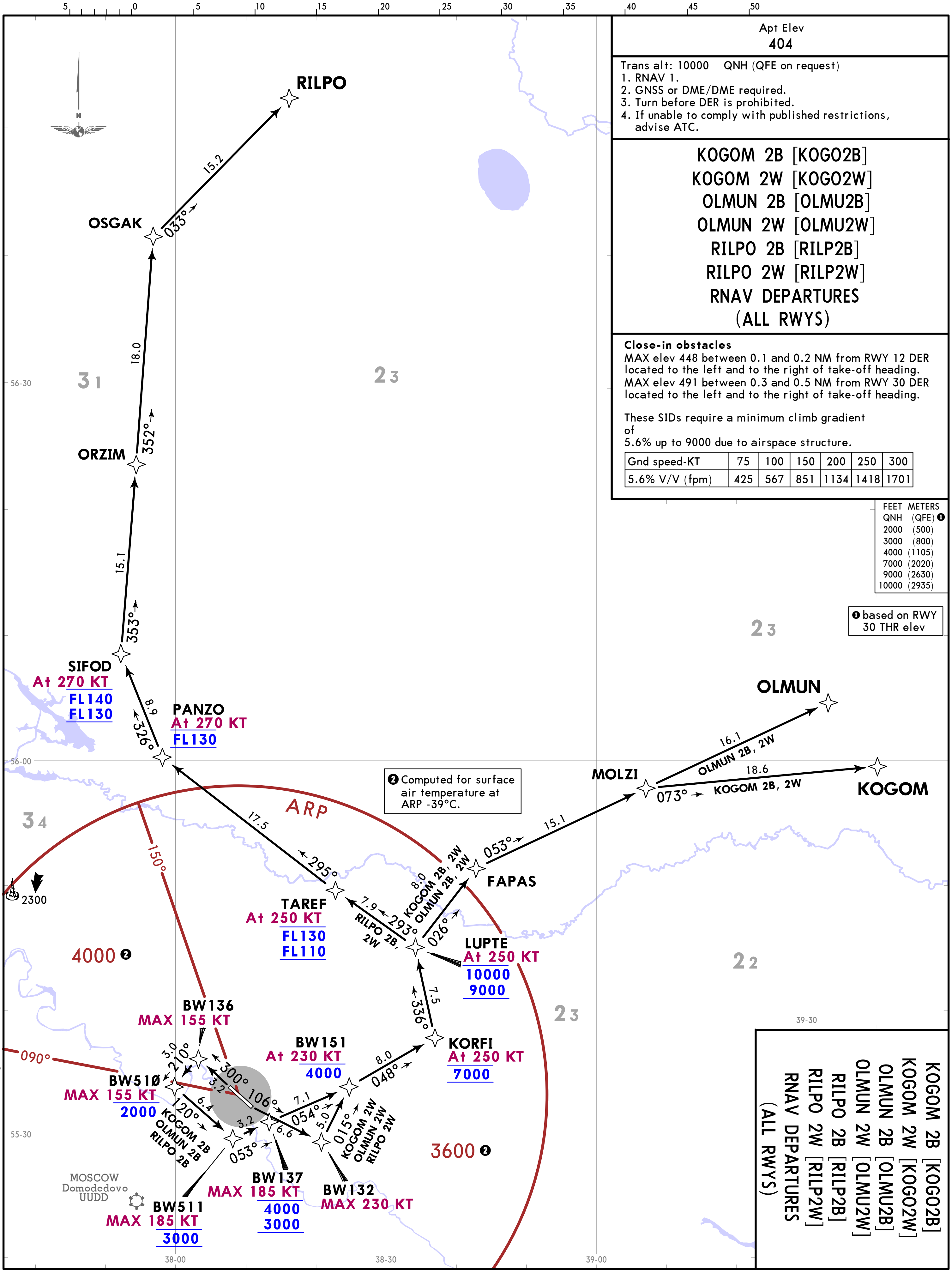
These SIDs require a minimum climb gradient of 5.6% up to 9000 due to airspace structure.

Gnd speed-KT	75	100	150	200	250	300
5.6% V/V (fpm)	425	567	851	1134	1418	1701

FEET QNH	METERS (QFE) ❶
2000	(500)
3000	(800)
4000	(1105)
7000	(2020)
9000	(2630)
10000	(2935)

❶ based on RWY 30 THR elev

❷ Computed for surface air temperature at ARP -39°C.



CHANGES: MSA raised.

29 SEP 23 10-3 Eff 5 Oct

JEPPESEN

RAMENSKOYE, RUSSIA
RNAV SID
KOGOM 2B [KOGO2B]
KOGOM 2W [KOGO2W]
OLMUN 2B [OLMU2B]
OLMUN 2W [OLMU2W]
RILPO 2B [RILP2B]
RILPO 2W [RILP2W]
RNAV DEPARTURES
(ALL RWYS)

CHANGES: MSA raised.

Apt Elev 404

Trans alt: 10000 QNH (QFE on request)

1. RNAV 1.
2. GNSS or DME/DME required.
3. Turn before DER is prohibited.
4. If unable to comply with published restrictions, advise ATC.

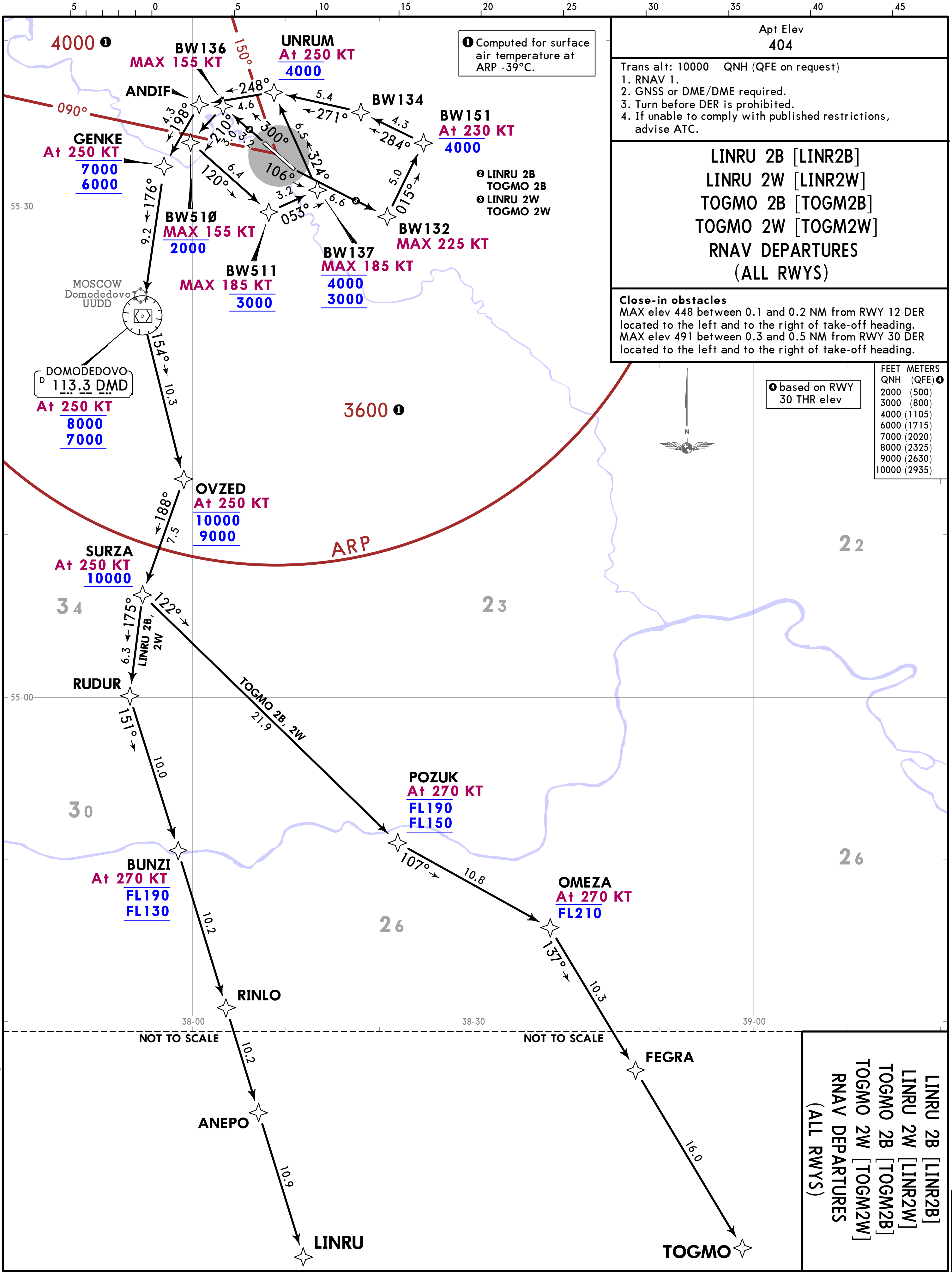
LINRU 2B [LINR2B]
LINRU 2W [LINR2W]
TOGMO 2B [TOGM2B]
TOGMO 2W [TOGM2W]
RNAV DEPARTURES
(ALL RWYS)

Close-in obstacles
MAX elev 448 between 0.1 and 0.2 NM from RWY 12 DER located to the left and to the right of take-off heading.
MAX elev 491 between 0.3 and 0.5 NM from RWY 30 DER located to the left and to the right of take-off heading.

FEET QNH	METERS (QFE) ❹
2000	(500)
3000	(800)
4000	(1105)
6000	(1715)
7000	(2020)
8000	(2325)
9000	(2630)
10000	(2935)

❹ based on RWY 30 THR elev

❶ Computed for surface air temperature at ARP -39°C.



Apt Elev
404

Trans alt: 10000 QNH (QFE on request)
1. RNAV 1.
2. GNSS or DME/DME required.
3. Turn before DER is prohibited.
4. If unable to comply with published restrictions, advise ATC.

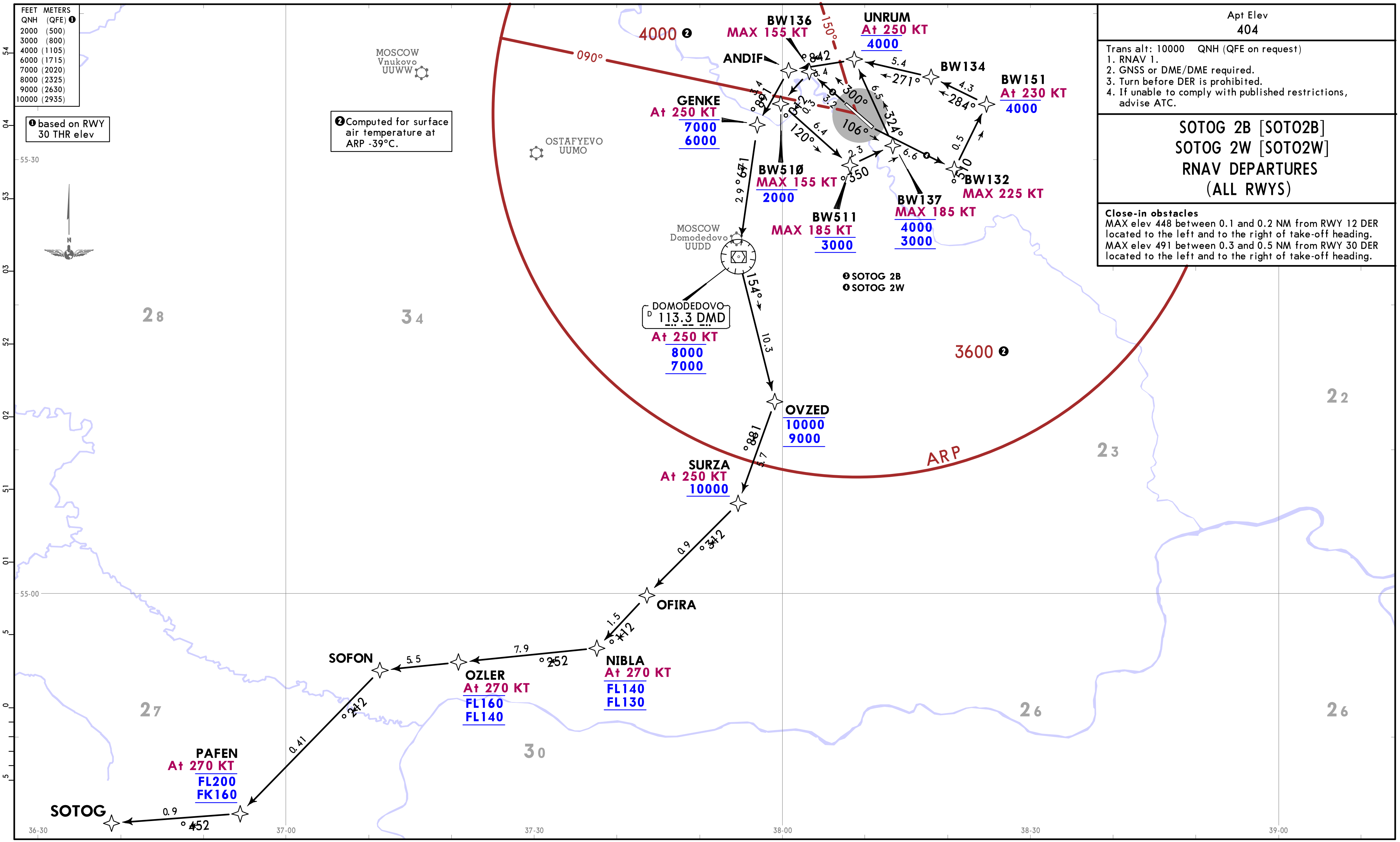
SOTOG 2B [SOTO2B]
SOTOG 2W [SOTO2W]
RNAV DEPARTURES
(ALL RWYS)

Close-in obstacles
MAX elev 448 between 0.1 and 0.2 NM from RWY 12 DER located to the left and to the right of take-off heading.
MAX elev 491 between 0.3 and 0.5 NM from RWY 30 DER located to the left and to the right of take-off heading.

FEET QNH	METERS (QFE) ❶
2000	(500)
3000	(800)
4000	(1105)
6000	(1715)
7000	(2020)
8000	(2325)
9000	(2630)
10000	(2935)

❶ based on RWY 30 THR elev

❷ Computed for surface air temperature at ARP -39°C.



FEET QNH	METERS (QFE) ❶
2000	(500)
3000	(800)
4000	(1105)
6000	(1715)
7000	(2020)
8000	(2325)
9000	(2630)
10000	(2935)

❶ based on RWY 30 THR elev

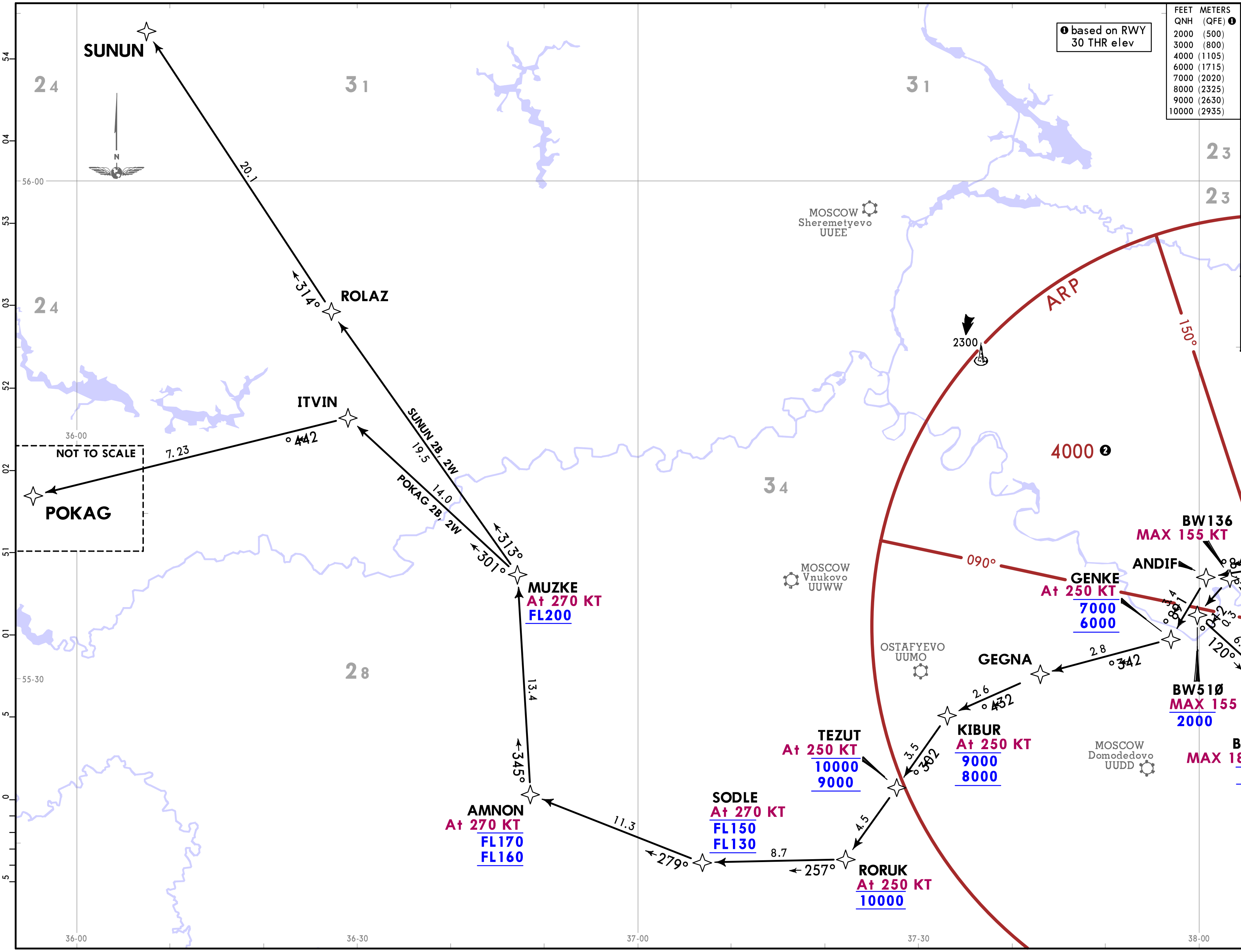
Apt Elev
404

Trans alt: 10000 QNH (QFE on request)
1. RNAV 1.
2. GNSS or DME/DME required.
3. Turn before DER is prohibited.
4. If unable to comply with published restrictions, advise ATC.

POKAG 2B [POKA2B]
POKAG 2W [POKA2W]
SUNUN 2B [SUNU2B]
SUNUN 2W [SUNU2W]
RNAV DEPARTURES (ALL RWYS)

Close-in obstacles
MAX elev 448 between 0.1 and 0.2 NM from RWY 12 DER located to the left and to the right of take-off heading.
MAX elev 491 between 0.3 and 0.5 NM from RWY 30 DER located to the left and to the right of take-off heading.

❷ Computed for surface air temperature at ARP -39°C.



Apt Elev 404	Trans alt: 10000 (QFE on request) 1. DME required until BW151. 2. DME/DME or GNSS required after BW151. 3. RNAV 1 after BW151. 4. If unable to comply with published restrictions, advise ATS.

RILPO 2A [RILP2A] DEPARTURE (RWY 30)

Close-in Obstacles
MAX elevation 491 - between 0.3 NM and 0.5 NM to the LEFT and RIGHT of take-off heading.

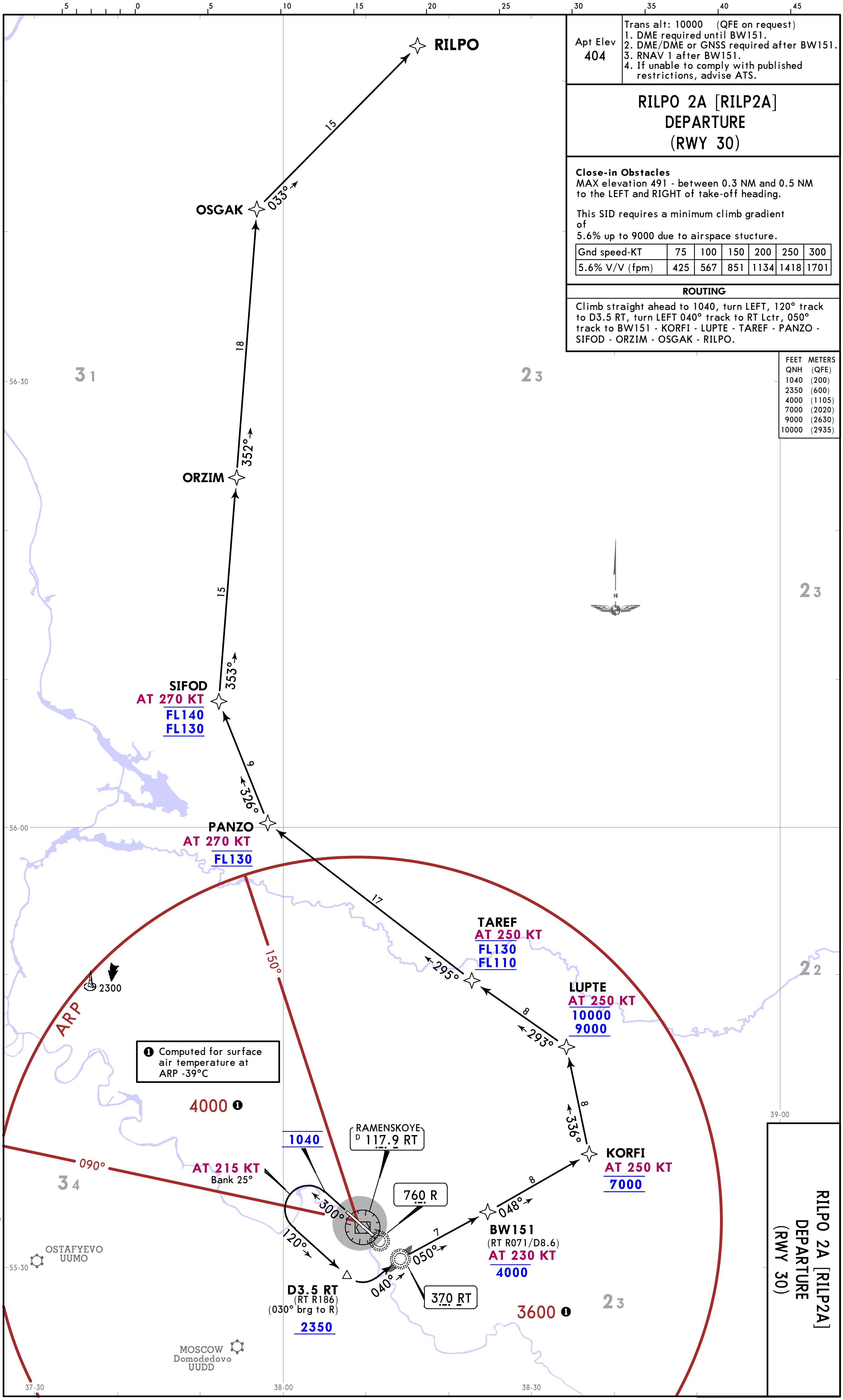
This SID requires a minimum climb gradient of 5.6% up to 9000 due to airspace stucture.

Gnd speed-KT	75	100	150	200	250	300
5.6% V/V (fpm)	425	567	851	1134	1418	1701

ROUTING

Climb straight ahead to 1040, turn LEFT, 120° track to D3.5 RT, turn LEFT 040° track to RT Lctr, 050° track to BW151 - KORFI - LUPTE - TAREF - PANZO - SIFOD - ORZIM - OSGAK - RILPO.

FEET QNH	METERS (QFE)
1040	(200)
2350	(600)
4000	(1105)
7000	(2020)
9000	(2630)
10000	(2935)



CHANGES: MSA raised.

CHANGES: MSA raised.

FEET QNH	METERS (QFE)
2700	(700)
4000	(1100)
7000	(2015)
9000	(2625)
10000	(2930)

Apt Elev
404

Trans alt: 10000 (QFE on request)
1. DME required until BW151.
2. DME/DME or GNSS required after BW151.
3. RNAV 1 after BW151.
4. If unable to comply with published restrictions, advise ATS.

RILPO 2X [RILP2X] DEPARTURE (RWY 12)

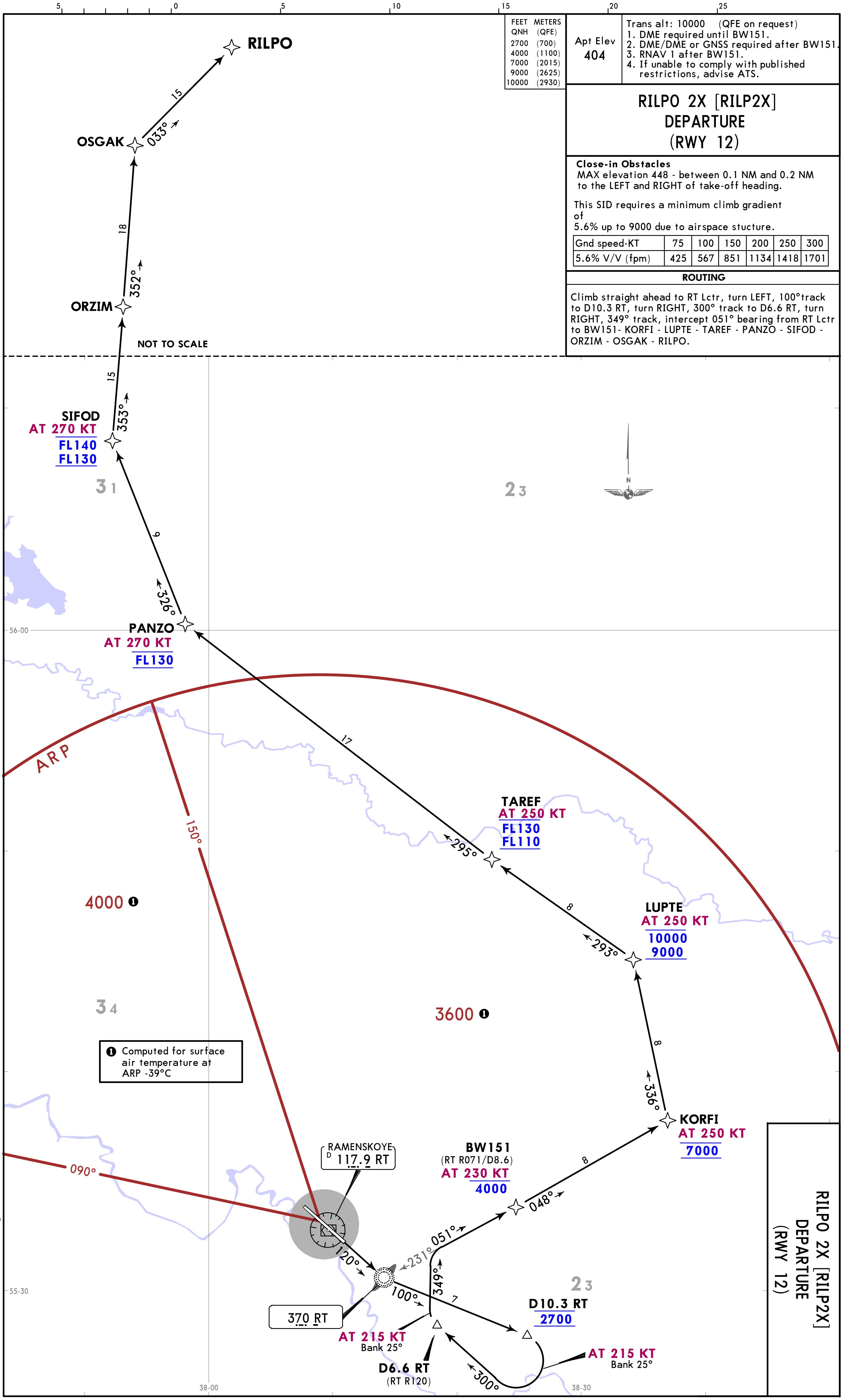
Close-in Obstacles
MAX elevation 448 - between 0.1 NM and 0.2 NM to the LEFT and RIGHT of take-off heading.

This SID requires a minimum climb gradient of
5.6% up to 9000 due to airspace stucture.

Gnd speed-KT	75	100	150	200	250	300
5.6% V/V (fpm)	425	567	851	1134	1418	1701

ROUTING

Climb straight ahead to RT Lctr, turn LEFT, 100°track to D10.3 RT, turn RIGHT, 300° track to D6.6 RT, turn RIGHT, 349° track, intercept 051° bearing from RT Lctr to BW151- KORFI - LUPTE - TAREF - PANZO - SIFOD - ORZIM - OSGAK - RILPO.

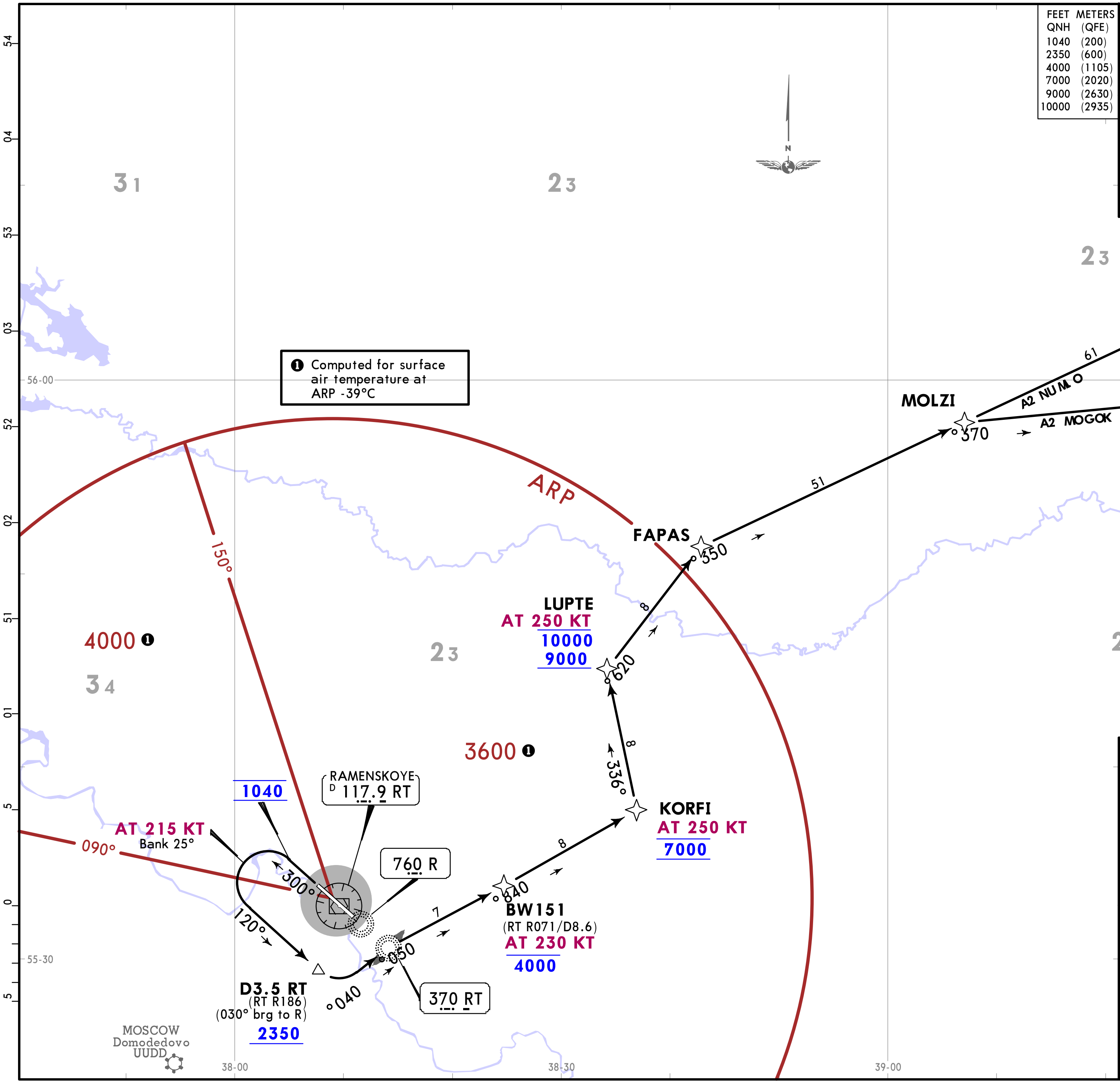


FEET QNH	METERS (QFE)
1040	(200)
2350	(600)
4000	(1105)
7000	(2020)
9000	(2630)
10000	(2935)

Apt Elev 404

Trans alt: 10000 (QFE on request)
1. DME required until BW151.
2. DME/DME or GNSS required after BW151.
3. RNAV 1 after BW151.
4. If unable to comply with published restrictions, advise ATS.

KOGOM 2A [KOGO2A] OLMUN 2A [OLMU2A] DEPARTURES (RWY 30)



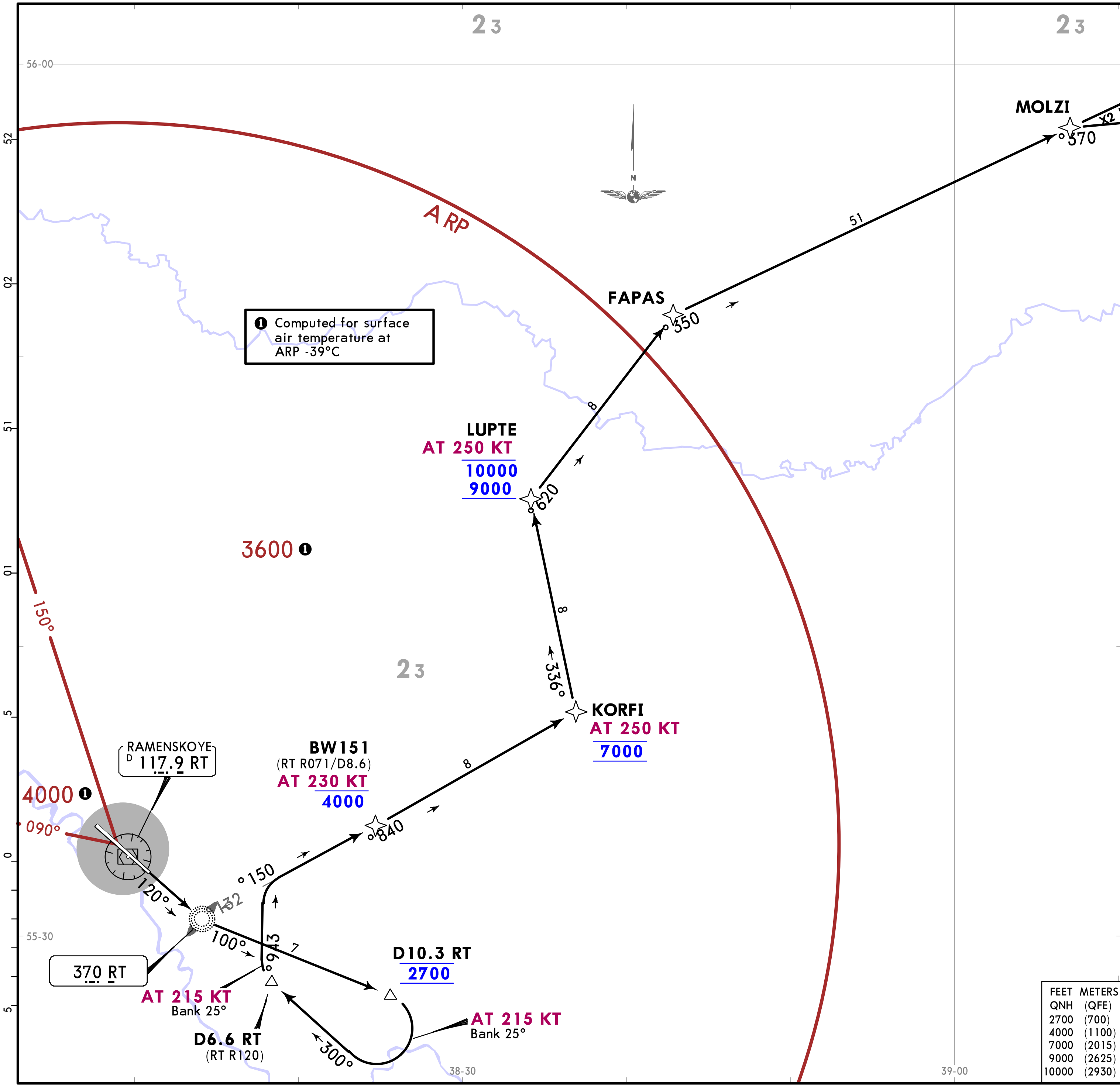
Close-in Obstacles
MAX elevation 491 - between 0.3 NM and 0.5 NM to the LEFT and RIGHT of take-off heading.

These SIDs require a minimum climb gradient of
5.6% up to 9000 due to airspace stucture.

Gnd speed-KT	75	100	150	200	250	300
5.6% V/V (fpm)	425	567	851	1134	1418	1701

SID	ROUTING
KOGOM 2A	Climb straight ahead to 1040, turn LEFT, 120° track to D3.5 RT, turn LEFT 040° track to RT Lctr, 050° track to BW151 - KORFI - LUPTE - FAPAS - MOLZI - KOGOM.
OLMUN 2A	Climb straight ahead to 1040, turn LEFT, 120° track to D3.5 RT, turn LEFT 040° track to RT Lctr, 050° track to BW151 - KORFI - LUPTE - FAPAS - MOLZI - OLMUN.

CHANGES: MSA raised.



FEET QNH	METERS (QFE)
2700	(700)
4000	(1100)
7000	(2015)
9000	(2625)
10000	(2930)

Apt Elev 404	Trans alt: 10000 (QFE on request) 1. DME required until BW151. 2. DME/DME or GNSS required after BW151. 3. RNAV 1 after BW151. 4. If unable to comply with published restrictions, advise ATS.

KOGOM 2X [KOGO2X]
OLMUN 2X [OLMU2X]
DEPARTURES
(RWY 12)

Close-in Obstacles
MAX elevation 448 - between 0.1 NM and 0.2 NM to the LEFT and RIGHT of take-off heading.

These SIDs require a minimum climb gradient of
5.6% up to 9000 due to airspace stucture.

Gnd speed-KT	75	100	150	200	250	300
5.6% V/V (fpm)	425	567	851	1134	1418	1701

SID	ROUTING
KOGOM 2X	Climb straight ahead to RT Lctr, turn LEFT, 100° track to D10.3 RT, turn RIGHT, 300° track to D6.6 RT, turn RIGHT, 349° track, intercept 051° bearing from RT Lctr to BW151 - KORFI - LUPTE - FAPAS - MOLZI - KOGOM.
OLMUN 2X	Climb straight ahead to RT Lctr, turn LEFT, 100° track to D10.3 RT, turn RIGHT, 300° track to D6.6 RT, turn RIGHT, 349° track, intercept 051° bearing from RT Lctr to BW151 - KORFI - LUPTE - FAPAS - MOLZI - OLMUN.

CHANGES: MSA raised.

FEET QNH	METERS (QFE)
1040	(200)
2350	(600)
6000	(1715)
7000	(2020)
8000	(2325)
9000	(2630)
10000	(2935)

Apt Elev 404

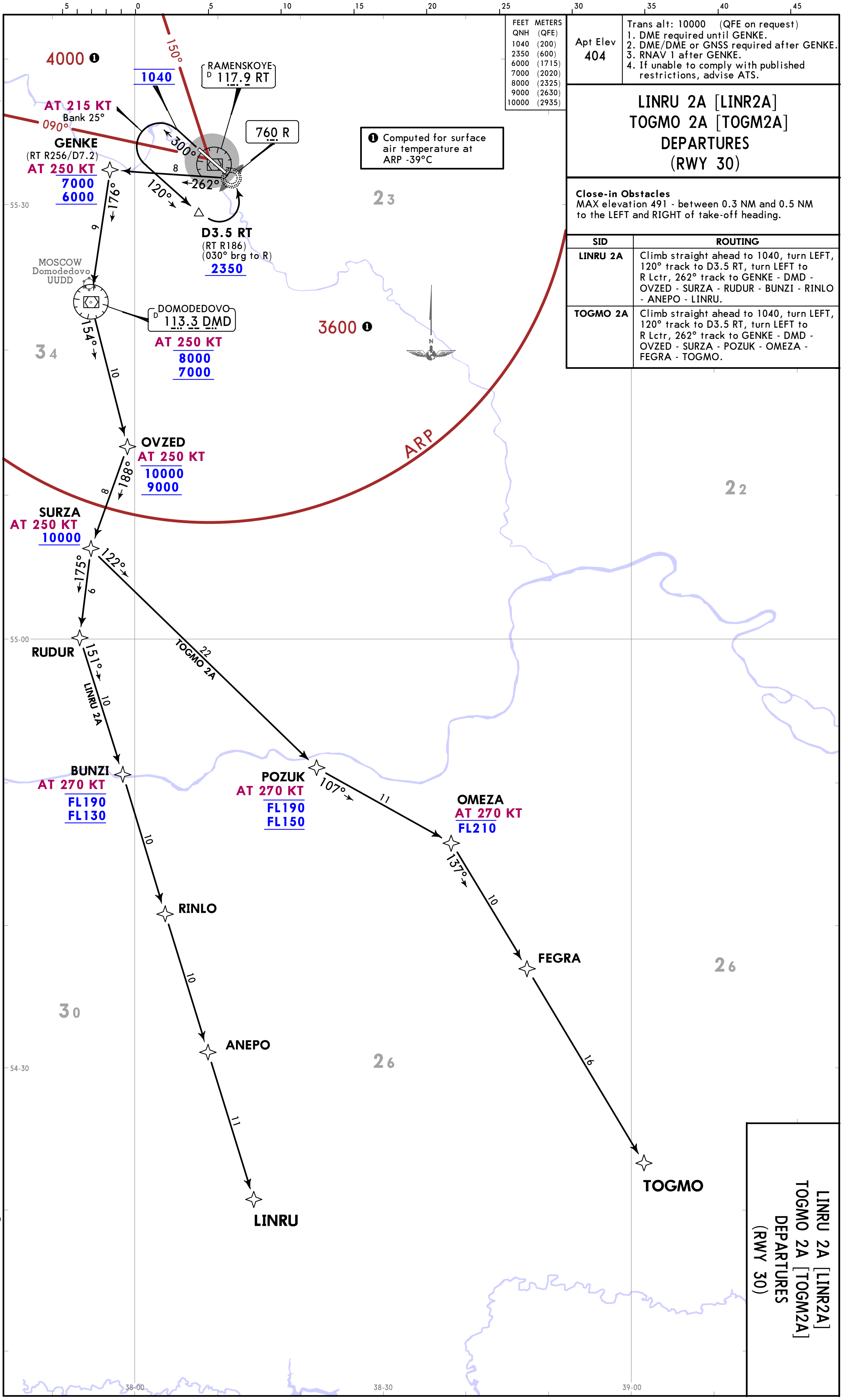
Trans alt: 10000 (QFE on request)
1. DME required until GENKE.
2. DME/DME or GNSS required after GENKE.
3. RNAV 1 after GENKE.
4. If unable to comply with published restrictions, advise ATS.

LINRU 2A [LINR2A] TOGMO 2A [TOGM2A] DEPARTURES (RWY 30)

Close-in Obstacles
MAX elevation 491 - between 0.3 NM and 0.5 NM to the LEFT and RIGHT of take-off heading.

SID	ROUTING
LINRU 2A	Climb straight ahead to 1040, turn LEFT, 120° track to D3.5 RT, turn LEFT to R Lctr, 262° track to GENKE - DMD - OVZED - SURZA - RUDUR - BUNZI - RINLO - ANEPO - LINRU.
TOGMO 2A	Climb straight ahead to 1040, turn LEFT, 120° track to D3.5 RT, turn LEFT to R Lctr, 262° track to GENKE - DMD - OVZED - SURZA - POZUK - OMEZA - FEGRA - TOGMO.

1 Computed for surface air temperature at ARP -39°C



LINRU 2A [LINR2A]
TOGMO 2A [TOGM2A]
DEPARTURES
(RWY 30)

CHANGES: MSA raised.

FEET QNH	METERS (QFE)
2700	(700)
6000	(1710)
7000	(2015)
8000	(2320)
9000	(2625)
10000	(2930)

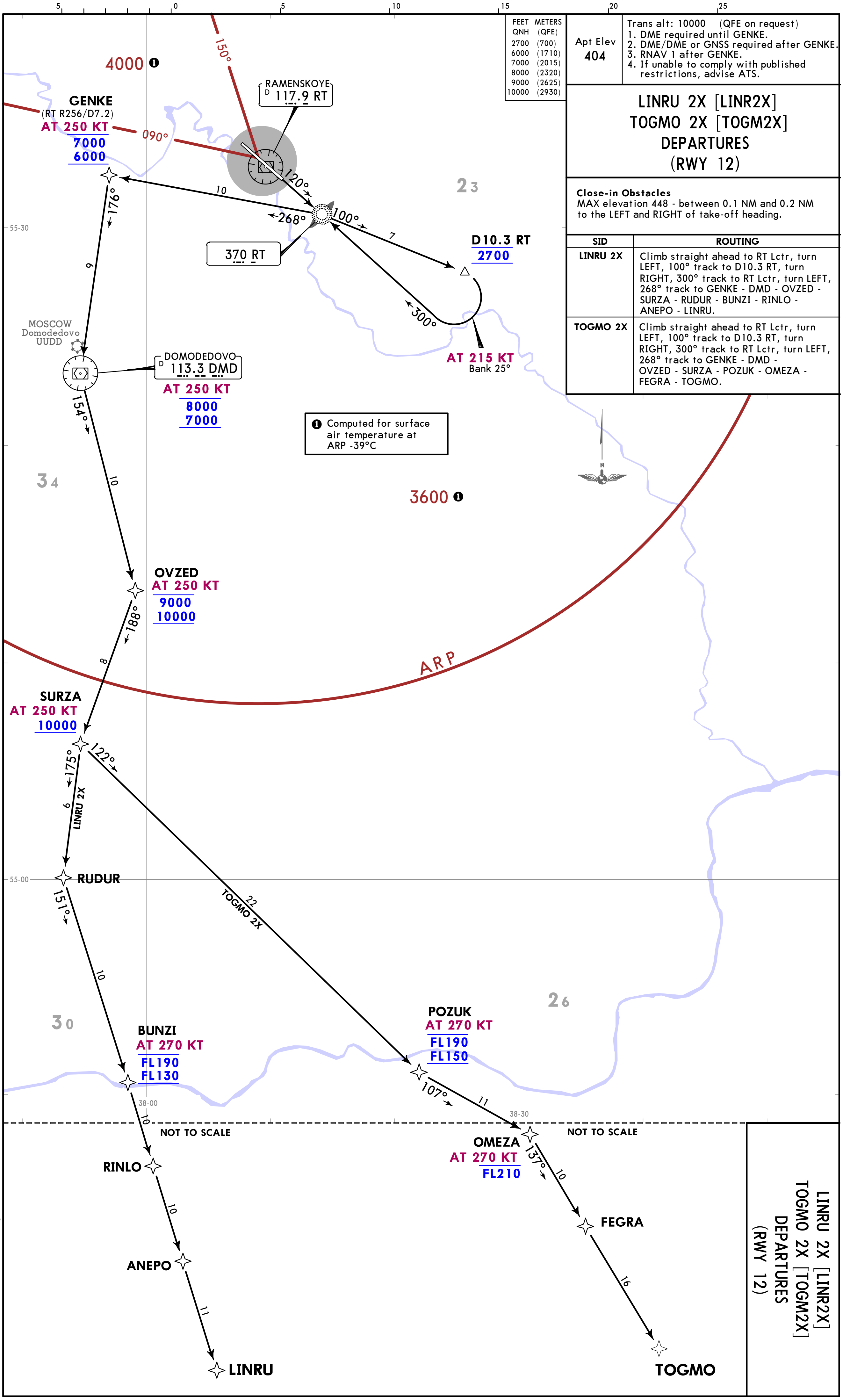
Apt Elev
404

Trans alt: 10000 (QFE on request)
1. DME required until GENKE.
2. DME/DME or GNSS required after GENKE.
3. RNAV 1 after GENKE.
4. If unable to comply with published restrictions, advise ATS.

LINRU 2X [LINR2X]
TOGMO 2X [TOGM2X]
DEPARTURES
(RWY 12)

Close-in Obstacles
MAX elevation 448 - between 0.1 NM and 0.2 NM to the LEFT and RIGHT of take-off heading.

SID	ROUTING
LINRU 2X	Climb straight ahead to RT Lctr, turn LEFT, 100° track to D10.3 RT, turn RIGHT, 300° track to RT Lctr, turn LEFT, 268° track to GENKE - DMD - OVZED - SURZA - RUDUR - BUNZI - RINLO - ANEPO - LINRU.
TOGMO 2X	Climb straight ahead to RT Lctr, turn LEFT, 100° track to D10.3 RT, turn RIGHT, 300° track to RT Lctr, turn LEFT, 268° track to GENKE - DMD - OVZED - SURZA - POZUK - OMEZA - FEGRA - TOGMO.



CHANGES: MSA raised.

Apt Elev 404	Trans alt: 10000 (QFE on request) 1. DME required until GENKE. 2. DME/DME or GNSS required after GENKE. 3. RNAV 1 after GENKE. 4. If unable to comply with published restrictions, advise ATS.

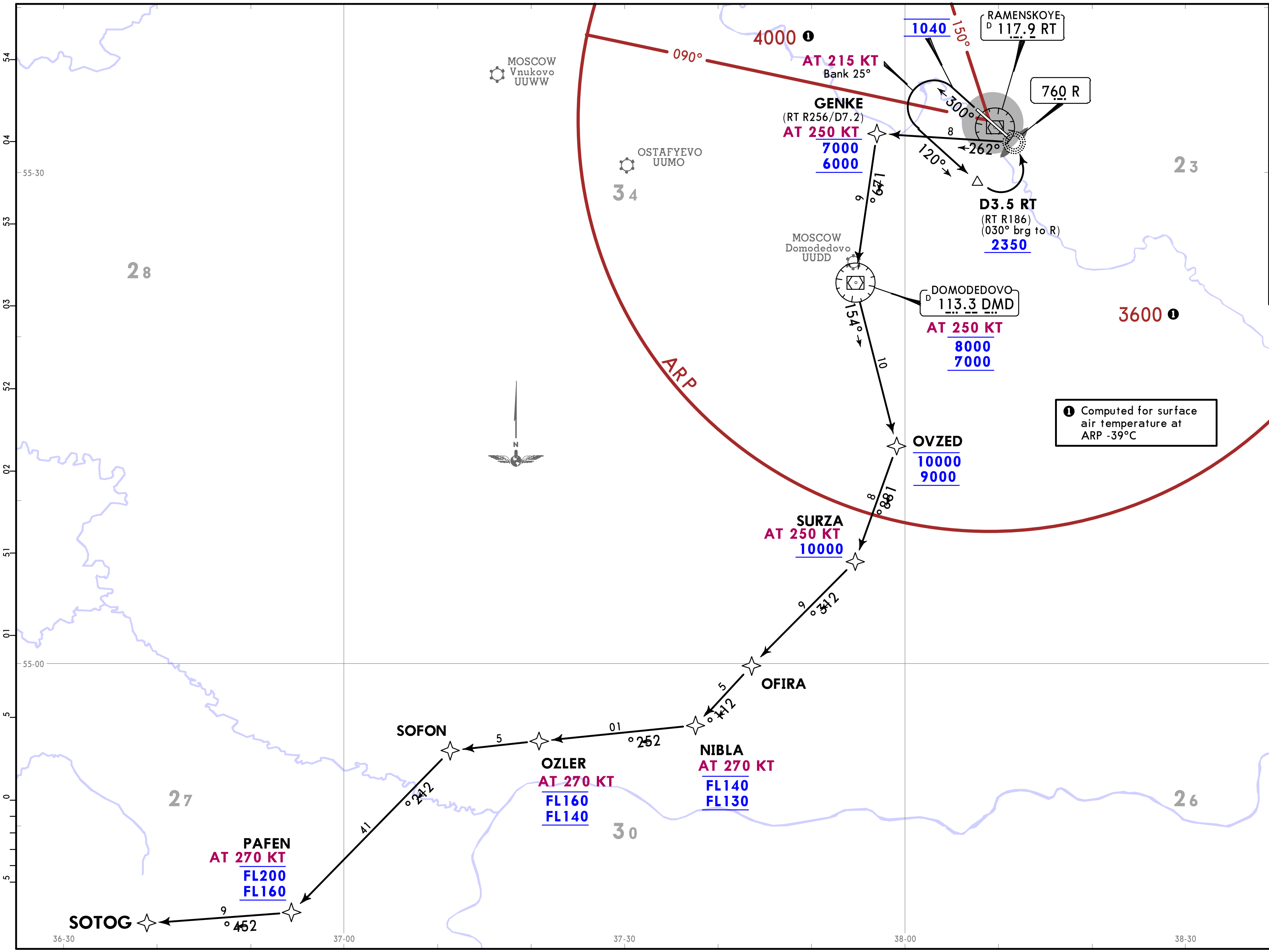
SOTOG 2A [SOTO2A] DEPARTURE (RWY 30)

Close-in Obstacles
MAX elevation 491 - between 0.3 NM and 0.5 NM to the LEFT and RIGHT of take-off heading.

ROUTING

Climb straight ahead to 1040, turn LEFT, 120° track to D3.5 RT, turn LEFT to R Lctr, 262° track to GENKE - DMD - OVZED - SURZA - OFIRA - NIBLA - OZLER - SOFON - PAFEN - SOTOG.

FEET QNH	METERS (QFE)
1040	(200)
2350	(600)
6000	(1715)
7000	(2020)
8000	(2325)
9000	(2630)
10000	(2935)



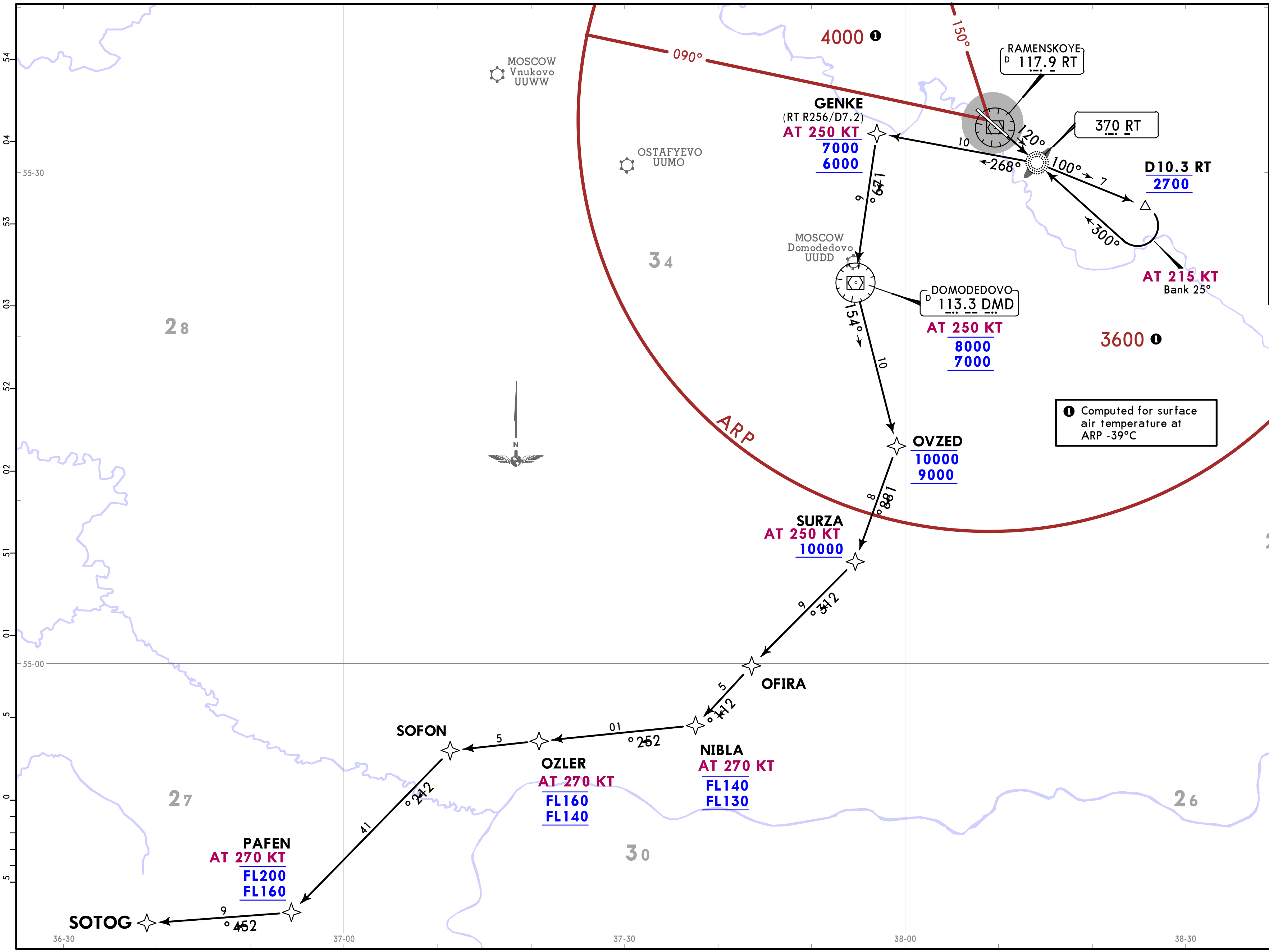
Apt Elev 404	Trans alt: 10000 (QFE on request) 1. DME required until GENKE. 2. DME/DME or GNSS required after GENKE. 3. RNAV 1 after GENKE. 4. If unable to comply with published restrictions, advise ATS.

SOTOG 2X [SOTO2X] DEPARTURE (RWY 12)

Close-in Obstacles
MAX elevation 448 - between 0.1 NM and 0.2 NM to the LEFT and RIGHT of take-off heading.

ROUTING
Climb straight ahead to RT Lctr, turn LEFT, 100° track to D10.3 RT, turn RIGHT, 300° track to RT Lctr, turn LEFT, 268° track to GENKE - DMD - OVZED - SURZA - OFIRA - NIBLA - OZLER - SOFON - PAFEN - SOTOG.

FEET QNH	METERS (QFE)
2700	(700)
6000	(1710)
7000	(2015)
8000	(2320)
9000	(2625)
10000	(2930)



❶ Computed for surface air temperature at ARP -39°C

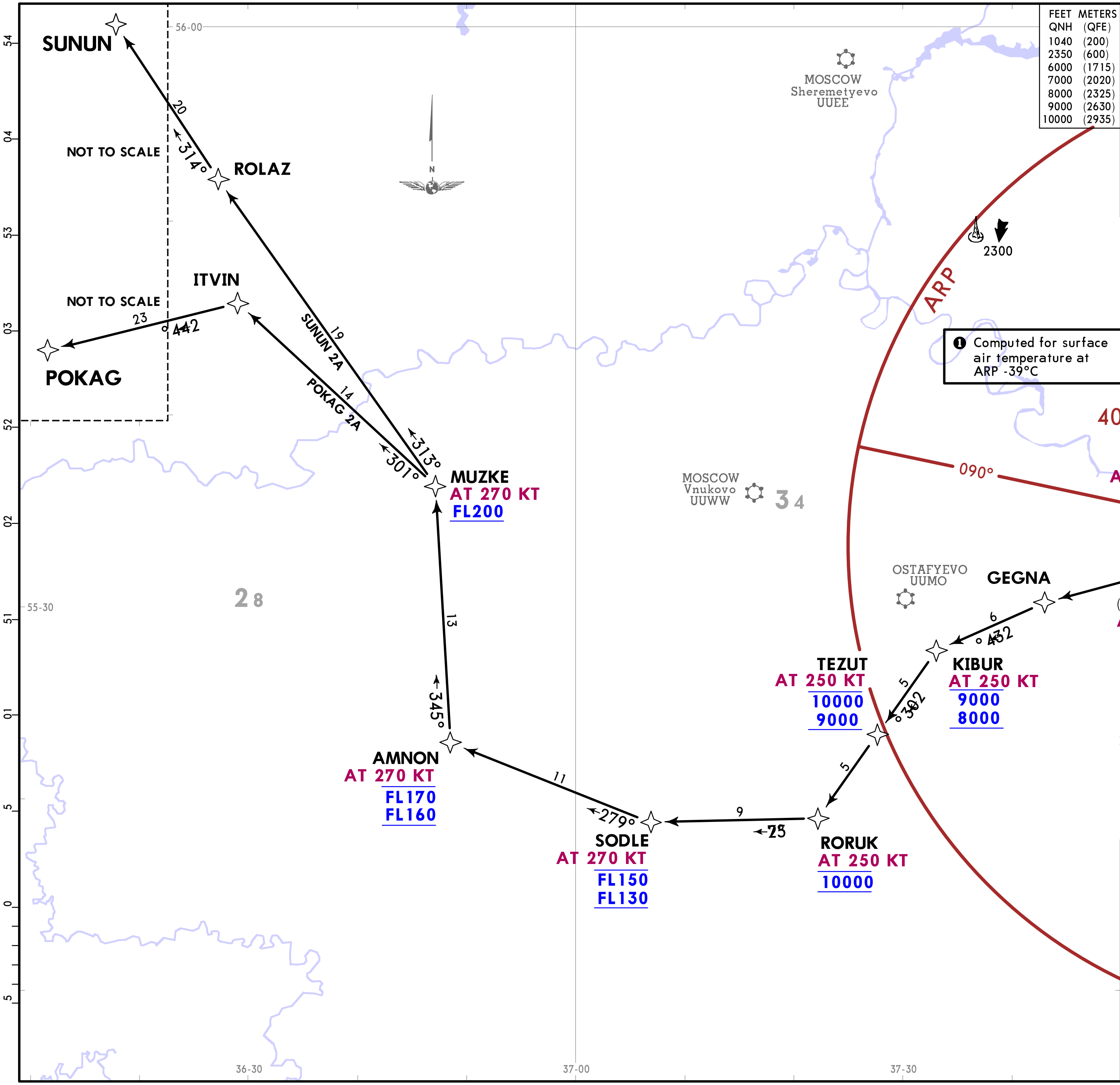
CHANGES: MSA raised.

FEET QNH	METERS (QFE)
1040	(200)
2350	(600)
6000	(1715)
7000	(2020)
8000	(2325)
9000	(2630)
10000	(2935)

Apt Elev
404

Trans alt: 10000 (QFE on request)
1. DME required until GENKE.
2. DME/DME or GNSS required after GENKE.
3. RNAV 1 after GENKE.
4. If unable to comply with published restrictions, advise ATS.

POKAG 2A [POKA2A]
SUNUN 2A [SUNU2A]
DEPARTURES
(RWY 30)



Close-in Obstacles
MAX elevation 491 - between 0.3 NM and 0.5 NM to the LEFT and RIGHT of take-off heading.

SID	ROUTING
POKAG 2A	Climb straight ahead to 1040, turn LEFT, 120° track to D3.5 RT, turn LEFT to R Lctr, 262° track to GENKE - GEGNA - KIBUR - TEZUT - RORUK - SODLE - AMNON - MUZKE - ITVIN - POKAG.
SUNUN 2A	Climb straight ahead to 1040, turn LEFT, 120° track to D3.5 RT, turn LEFT to R Lctr, 262° track to GENKE - GEGNA - KIBUR - TEZUT - RORUK - SODLE - AMNON - MUZKE - ROLAZ - SUNUN.

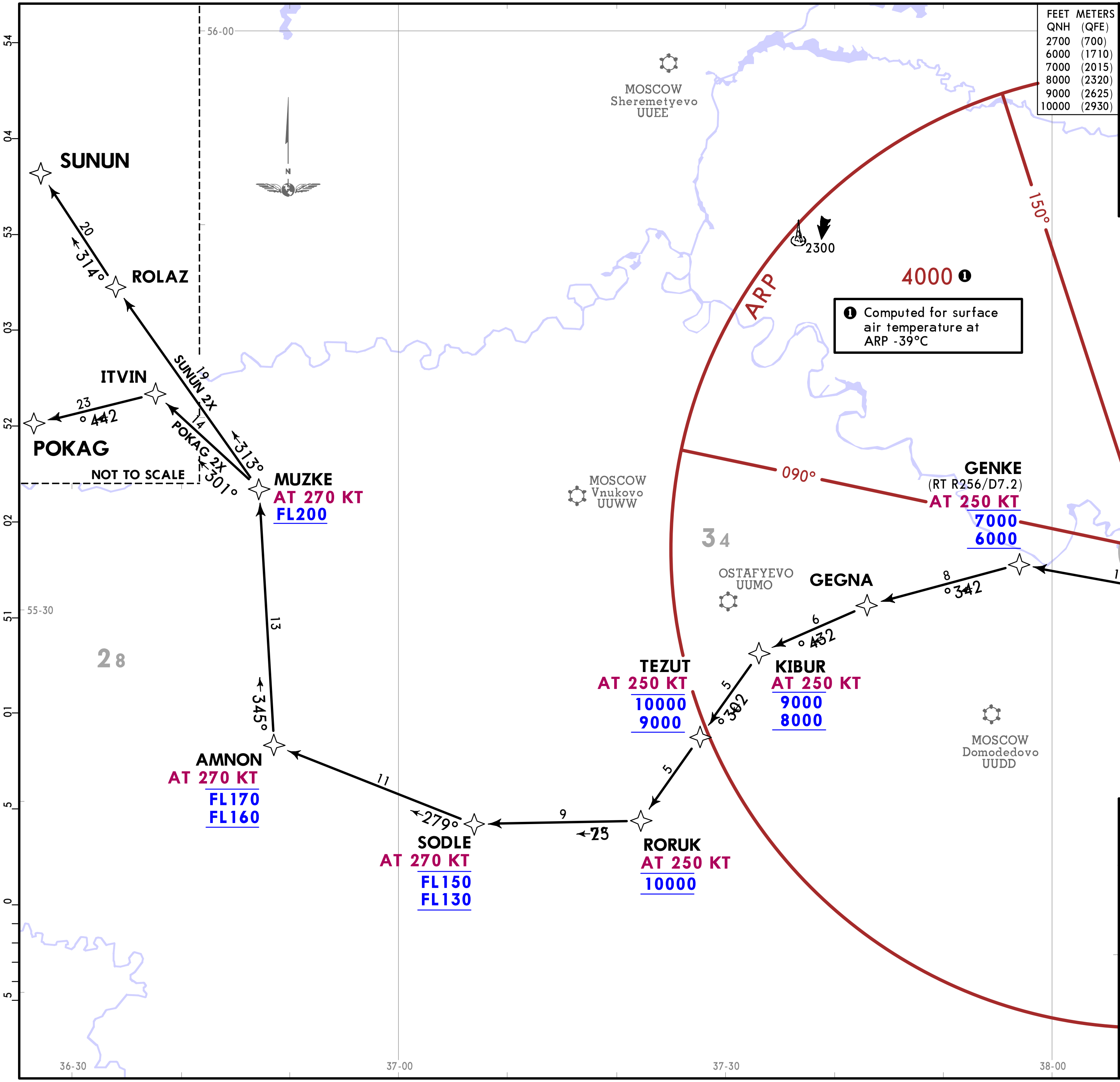
CHANGES: MSA raised.

FEET QNH	METERS (QFE)
2700	(700)
6000	(1710)
7000	(2015)
8000	(2320)
9000	(2625)
10000	(2930)

Apt Elev
404

Trans alt: 10000 (QFE on request)
1. DME required until GENKE.
2. DME/DME or GNSS required after GENKE.
3. RNAV 1 after GENKE.
4. If unable to comply with published restrictions, advise ATS.

POKAG 2X [POKA2X]
SUNUN 2X [SUNU2X]
DEPARTURES
(RWY 12)



Close-in Obstacles
MAX elevation 448 - between 0.1 NM and 0.2 NM to the LEFT and RIGHT of take-off heading.

SID	ROUTING
POKAG 2X	Climb straight ahead to RT Lctr, turn LEFT, 100° track to D10.3 RT, turn RIGHT, 300° track to RT Lctr, turn LEFT, 268° track to GENKE - GEGNA - KIBUR - TEZUT - RORUK - SODLE - AMNON - MUZKE - ITVIN - POKAG.
SUNUN 2X	Climb straight ahead to RT Lctr, turn LEFT, 100° track to D10.3 RT, turn RIGHT, 300° track to RT Lctr, turn LEFT, 268° track to GENKE - GEGNA - KIBUR - TEZUT - RORUK - SODLE - AMNON - MUZKE - ROLAZ - SUNUN.

CHANGES: MSA raised.

Apt Elev 404	Trans alt: 10000 QNH (QFE on request)

FLIGHT ROUTES BETWEEN AERODROMES WITHIN MOSCOW TMA

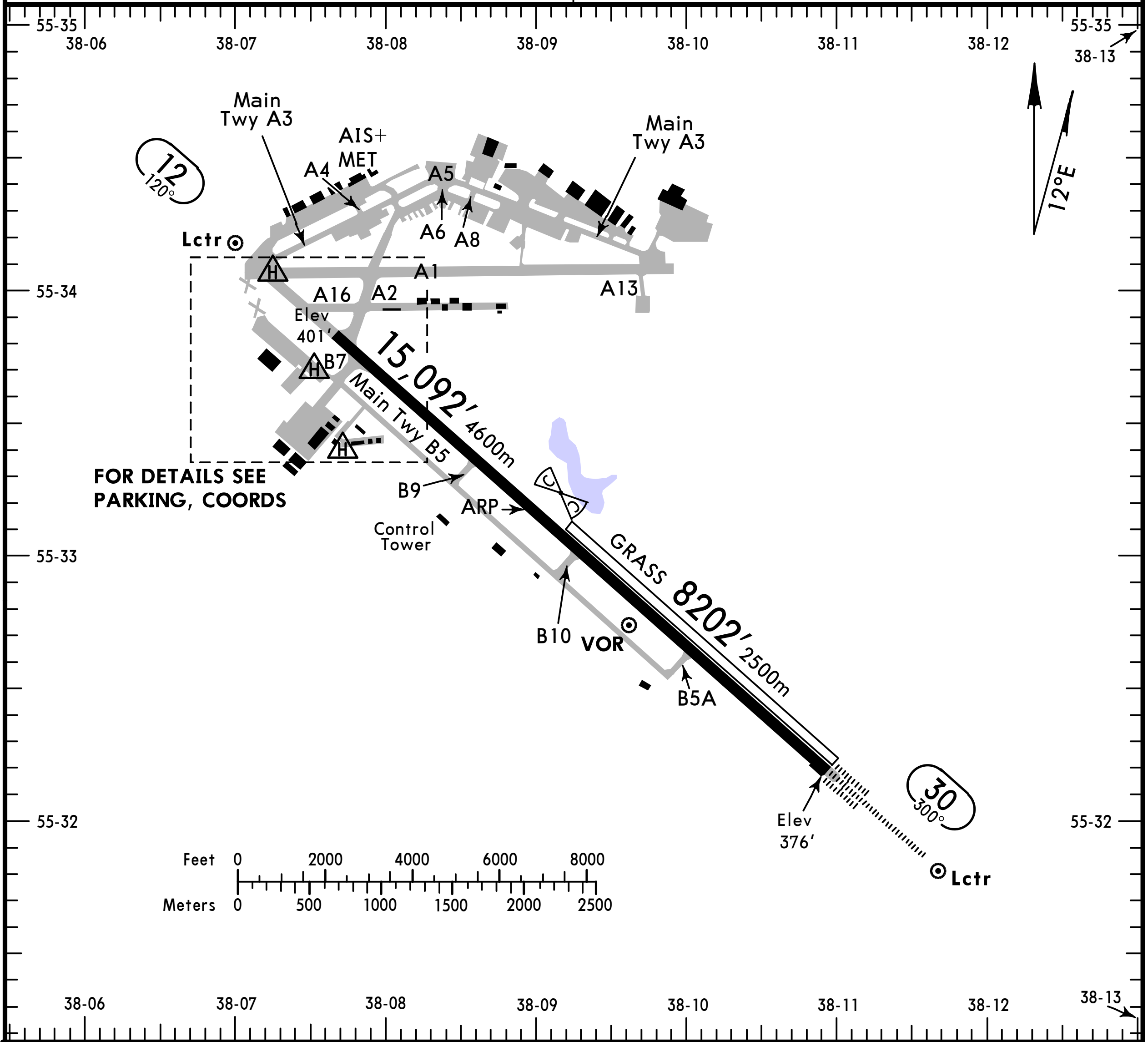
1. Departure instructions provide ACFT vectoring to the significant point on the route (the first waypoint in the flight plan);
2. Flights within CTRs shall be carried out via waypoints, separated by letters DCT in the flight plan, to IAF of the destination aerodrome (in accordance with the information published for appropriate departure aerodrome);
3. Approach shall be executed from IAF of the destination aerodrome:
   - Moscow/Sheremetyevo - KEZVU (IAF)
   - Moscow/Domodedovo - ALBOR (IAF)
   - Moscow/Vnukovo - RORUK (IAF)
   - Ostafyevo - RORUK (IAF)
   - Ramenskoye - ODLOR (IAF).

Departure To	ROUTING
Moscow/ Sheremetyevo	ANDIF - GEKLA - RUGEL - BESTA - SORET - RIMDE - KN NDB - EEØ43 - EEØ44 - AGMER - EEØ45 - TAFAZ - KEZVU (IAF).
Moscow/ Domodedovo	ANDIF - IMZUP - KUPVE - NIDBE - IZVOK - IPKED - ZOVGO - ODZAG - GUFUZ - ALBOR (IAF).
Moscow/ Vnukovo	ANDIF - IMZUP - GEGNA - KIBUR - LO NDB - BEMAS - TEBDI - TEPTA - RONEZ - TOLKE - TADUT - FIDOT - RORUK (IAF).
Ostafyevo	ANDIF - IMZUP - GEGNA - KIBUR - LO NDB - BEMAS - TEBDI - TEPTA - RONEZ - TOLKE - TADUT - FIDOT - RORUK (IAF).

FLIGHT INFORMATION SERVICE

In the absence of radio communication, flight crew may use mobile communication:
+7 (495) 556 - 55 - 49, ATS unit controller.

*GORDY Taxiing (GND)	*Tower
131.0	125.25



ADDITIONAL RUNWAY INFORMATION

RWY		USABLE LENGTHS LANDING BEYOND Threshold	Glide Slope	TAKE-OFF	WIDTH
12	HIRL (60m) CL (15m) PAPI-L (3.0°) RVR			❸	230' 70m
30	HIRL (60m) CL (15m) ❶HIALS-II TDZ ❷ RVR		13,999' 4267m		
12 / 30	Grass runway				197' 60m

❶ length 900m
❷ PAPI-L (angle 3.0°)
❸ TAKE-OFF RUN AVAILABLE

RWY 12:
From rwy head 15,092' (4600m)
twy B7 int 14,436' (4400m)
twy B9 int 10,827' (3300m)
twy B10 int 7677' (2340m)

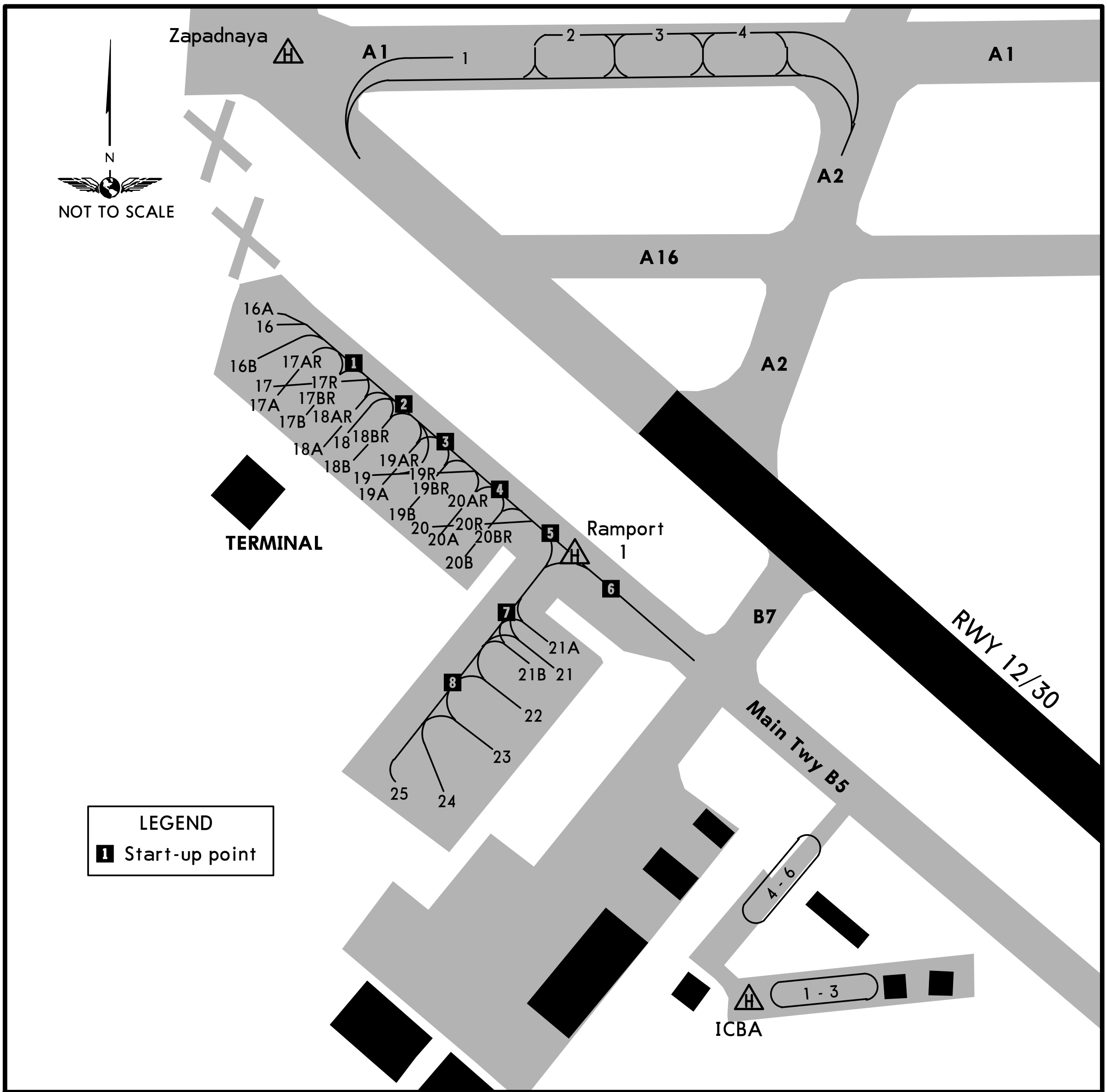
RWY 30:
From rwy head 15,092' (4600m)
twy B5A int 11,089' (3380m)
twy B10 int 7546' (2300m)
twy B9 int 4396' (1340m)

Std TAKE-OFF

HIRL & CL (spacing 15m or less) & relevant RVR	RL & CL & relevant RVR	RL & CL	❶ RL & RCLM	❶ RL or RCLM	Adequate Vis Ref DAY	Adequate Vis Ref NIGHT
TDZ R125m Mid R125m Rollout R125m	TDZ R150m Mid R150m Rollout R150m	R/V200m	R/V300m	R/V400m	R/V500m	NA

❶ For NIGHT operations, at least RL or CL and RENL are required.

CHANGES: Communications.



INS COORDINATES

STAND No.	COORDINATES	STAND No.	COORDINATES
	TWY A1		**APRON**
1	N55 34.1 E038 07.5	16 thru 16B	N55 33.9 E038 07.1
2	N55 34.1 E038 07.6	17	N55 33.8 E038 07.1
3	N55 34.1 E038 07.7	17A	N55 33.8 E038 07.2
4	N55 34.1 E038 07.8	17AR	N55 33.9 E038 07.2
		17B thru 18AR	N55 33.8 E038 07.2
		18B thru 19B	N55 33.8 E038 07.3
		19BR	N55 33.8 E038 07.4
		19R	N55 33.8 E038 07.3
		20 thru 20R	N55 33.7 E038 07.4
		21 thru 21B	N55 33.6 E038 07.5
		22	N55 33.6 E038 07.4
		23	N55 33.5 E038 07.4
		24, 25	N55 33.5 E038 07.3

CHANGES: None.

STRAIGHT-IN RWY		A	B	C	D
12	RNP LNAV/VNAV	**678'**(277') R1300m	**688'**(287') R1400m	**698'**(297') R1400m	**708'**(307') R1400m
	❶RNP LNAV	**800'**(399') R1500m	**800'**(399') R1500m	**800'**(399') R1800m	**800'**(399') R1800m
	❶ VOR X	**850'**(449') R1500m	**850'**(449') R1500m	**850'**(449') R2100m	**850'**(449') R2100m
	❶ NDB Y	**880'**(479') R1500m	**880'**(479') R1500m	**880'**(479') R2200m	**880'**(479') R2200m
	❶ NDB X	**860'**(459') R1500m	**860'**(459') R1500m	**860'**(459') R2100m	**860'**(459') R2100m
30	ILS FULL TDZ or CL out ALS out	**576'**(200') R550m ❷ R550m R1200m	**576'**(200') R550m ❷ R550m R1200m	**576'**(200') R550m ❷ R550m R1200m	**576'**(200') R550m ❷ R550m R1200m
	RNP LNAV/VNAV ALS out	**626'**(250') ❸ R750m R1300m	**626'**(250') ❸ R750m R1300m	**630'**(254') ❹ R750m R1300m	**650'**(274') ❹ R750m R1300m
	❶ RNP LNAV ALS out	**700'**(324') R800m R1500m	**700'**(324') R800m R1500m	**700'**(324') R800m R1500m	**700'**(324') R800m R1500m
	❶ VOR X ALS out	**690'**(314') R750m R1400m	**690'**(314') R750m R1400m	**690'**(314') R750m R1400m	**690'**(314') R750m R1400m
	❶ NDB Y or NDB X ALS out	**730'**(354') R900m R1500m	**730'**(354') R900m R1500m	**730'**(354') R900m R1600m	**730'**(354') R900m R1600m

❶ Continuous Descent Final Approach.
❷ RVR 750m when a Flight Director or Autopilot or HUD to DA is not used.
❸ With TDZ & CL & HUD: R550m
❹ With TDZ & CL & HUD: R600m

CIRCLE-TO-LAND	A	B	C	D
After VOR X or NDB X Rwy 12	**980'**(576') V1500m	**1040'**(636') V1600m	**1160'**(756') V2400m	**1160'**(756') V3600m

TAKE-OFF

Low Visibility Take-off					RL or RCLM	RL or CL	Adequate Vis Ref	
HIRL & CL (spacing 15m or less) & relevant RVR	RL & CL & relevant RVR	RL & CL	RL & RCLM	RL or CL				
			DAY	NIGHT	DAY	NIGHT	DAY	NIGHT
TDZ R125m Mid R125m Rollout R125m	TDZ R150m Mid R150m Rollout R150m	R200m	R300m		R/V400m		R/V500m	NA

UUBW/ZIA
RAMENSKOYE

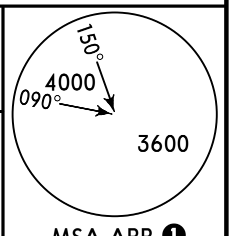
JEPPESEN
29 DEC 23 (11-1)

RAMENSKOYE, RUSSIA
ILS Rwy 30

BRIEFING STRIP ™

*GORDY Approach/Radar	*GORDY Tower	*Ground
125.25	125.25	131.0

LOC IRT	Final Apch Crs	BW333	DA(H)	Apt Elev 404′ Rwy 376′
*110.9	300°	2000′(1624′)	576′(200′)	

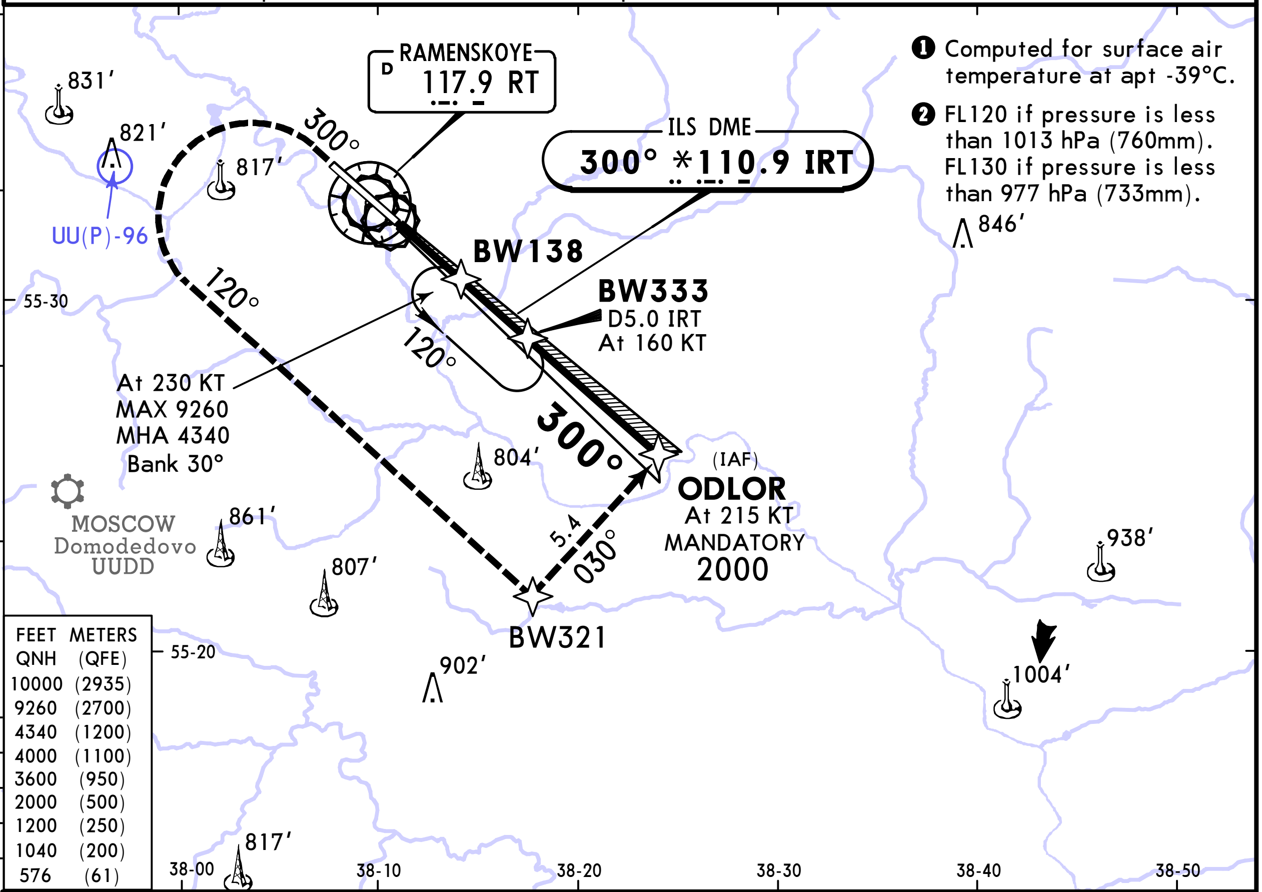


MSA ARP ❶

MISSED APCH: Climb STRAIGHT AHEAD to 1040′ or above (at 190 KT), then turn LEFT to BW321 (at 215 KT) climbing to 2000′, then proceed to ODLOR (at 215 KT) for instrument approach, or as directed.

Alt Set: hPa (MM on req) Rwy Elev: 14 hPa Trans level: FL110 ❷ Trans alt: 10000′

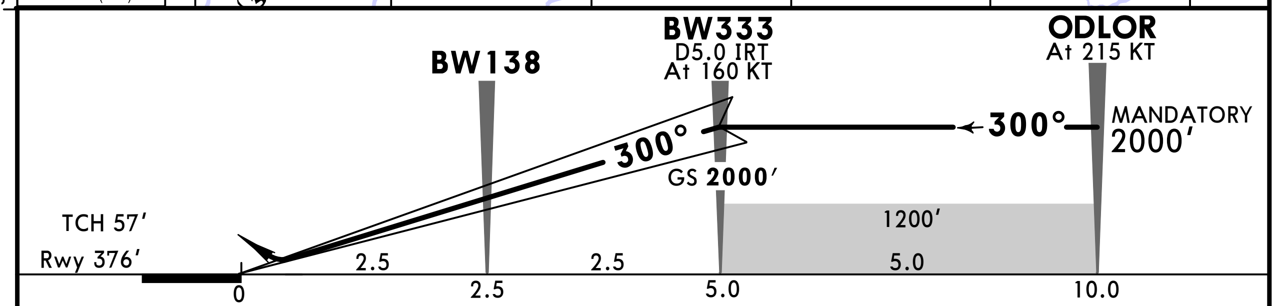
RNAV 1 for initial, intermediate & missed apch.
DME/DME or GNSS required for initial/intermediate apch.



❶ Computed for surface air temperature at apt -39°C.

❷ FL120 if pressure is less than 1013 hPa (760mm). FL130 if pressure is less than 977 hPa (733mm).

FEET QNH	METERS (QFE)
10000	(2935)
9260	(2700)
4340	(1200)
4000	(1100)
3600	(950)
2000	(500)
1200	(250)
1040	(200)
576	(61)



Gnd speed-Kts	70	90	100	120	140	160
GS 3.00°	372	478	531	637	743	849

HIALS-II
PAPI

MIN 1040′ | 190 KT

Std

STRAIGHT-IN LANDING
ILS
DA(H) **576′** (200′)

	Full	TDZ or CL out	ALS out
A			
B			
C	R550m	❶ R550m	R1200m
D			

❶ R750m when a Flight Director or Autopilot or HUD to DA is not used.

PANS OPS

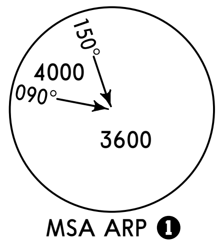
CHANGES: Communications.

BRIEFING STRIP ™

*GORDY Approach/Radar	*GORDY Tower	*Ground
125.250	125.250	131.0

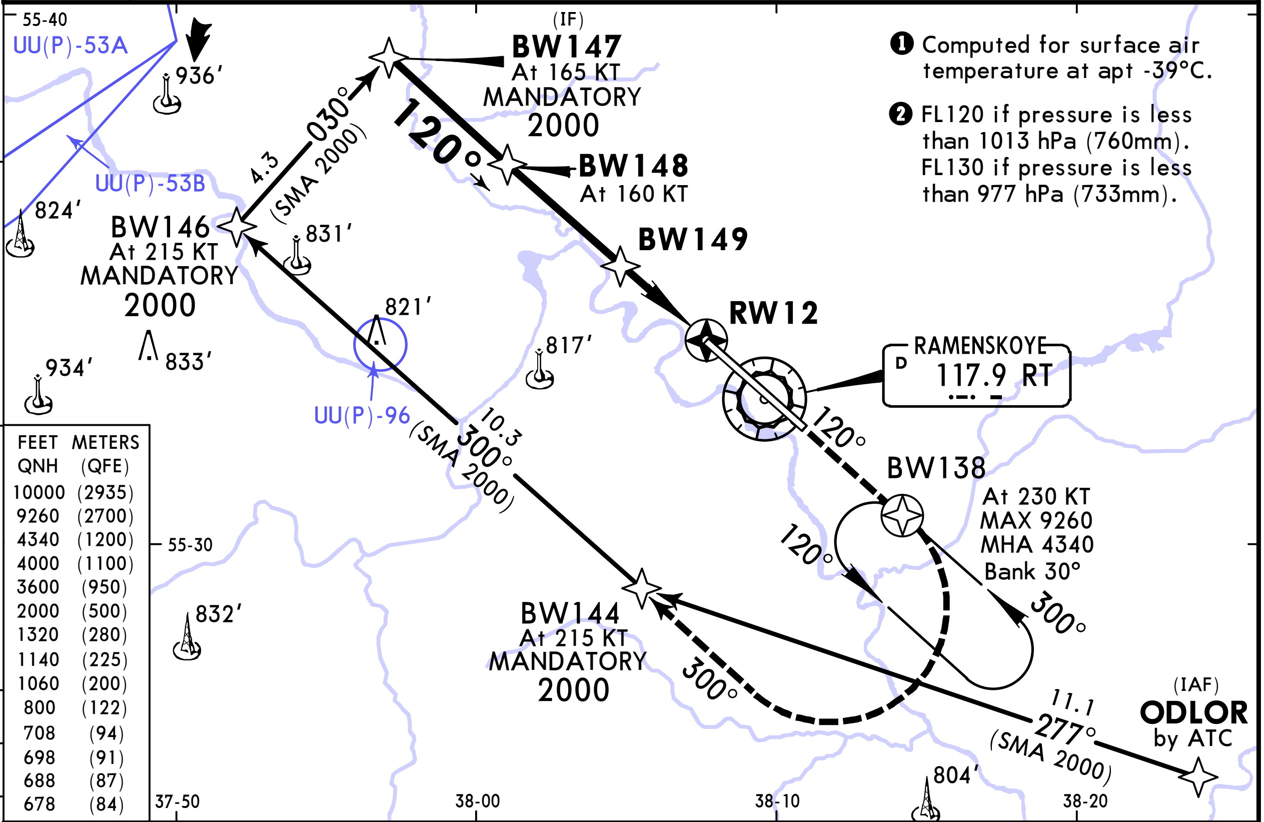
RNAV	Final Apch Crs 120°	BW148 MANDATORY 2000′ (1599′)	LNAV/VNAV DA(H) Refer to Minimums	Apt Elev 404′ Rwy 401′

MISSED APCH: Climb STRAIGHT AHEAD to 1060′ to BW138 (at 190 KT), then turn RIGHT to BW144 climbing to 2000′ (at 215 KT), then according to chart, or as directed.



Alt Set: hPa (MM on req)	Rwy Elev: 15 hPa	Trans level: FL110 ❷	Trans alt: 10000′

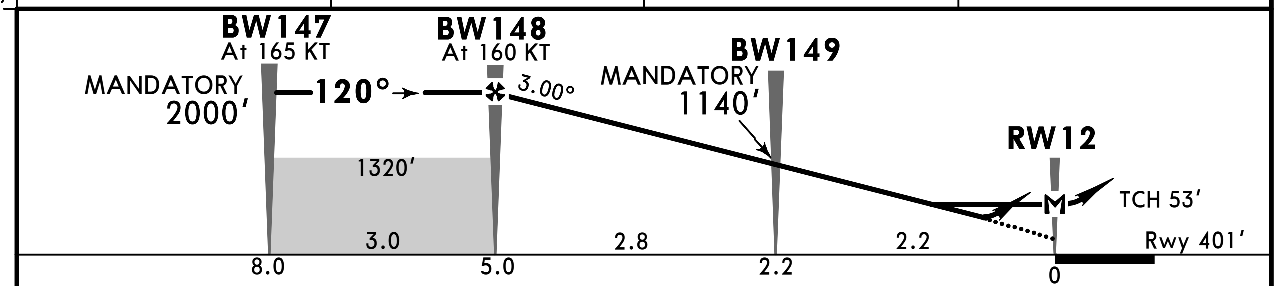
RNP apch.
1. GNSS required.
2. Baro-VNAV not authorized below -31°C.



FEET QNH	METERS (QFE)
10000	(2935)
9260	(2700)
4340	(1200)
4000	(1100)
3600	(950)
2000	(500)
1320	(280)
1140	(225)
1060	(200)
800	(122)
708	(94)
698	(91)
688	(87)
678	(84)

DIST to BW149	2.7	2.2	1.6	1.1	0.5
ALT	2000′	1830′	1670′	1490′	1320′

DIST to RW12	1.6	1.1	0.5
ALT	980′	800′	630′



Gnd speed-Kts		70	90	100	120	140	160
Glide Path Angle	3.00°	372	478	531	637	743	849
MAP at RW12							

PAPI-L	1060′ ↑	BW138	190 KT

Std

STRAIGHT-IN LANDING

	LNAV/VNAV DA(H) A: 678′ (277′) C: 698′ (297′) B: 688′ (287′) D: 708′ (307′)	LNAV CDFA ❶ DA/MDA(H) 800′ (399′)
A	R1300m	R1500m
B	R1400m	R1500m
C	R1400m	R1800m
D	R1400m	R1800m

PANS OPS

❶ VNAV DA(H) in lieu of MDA(H) depends on operator policy.

CHANGES: Note.

UUBW/ZIA RAMENSKOYE

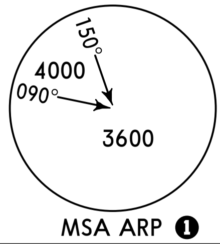
JEPPESEN
27 SEP 24 (12-2) Eff 3 Oct

RAMENSKOYE, RUSSIA RNP Rwy 30

BRIEFING STRIP ™

*GORDY Approach/Radar	*GORDY Tower	*Ground
125.250	125.250	131.0

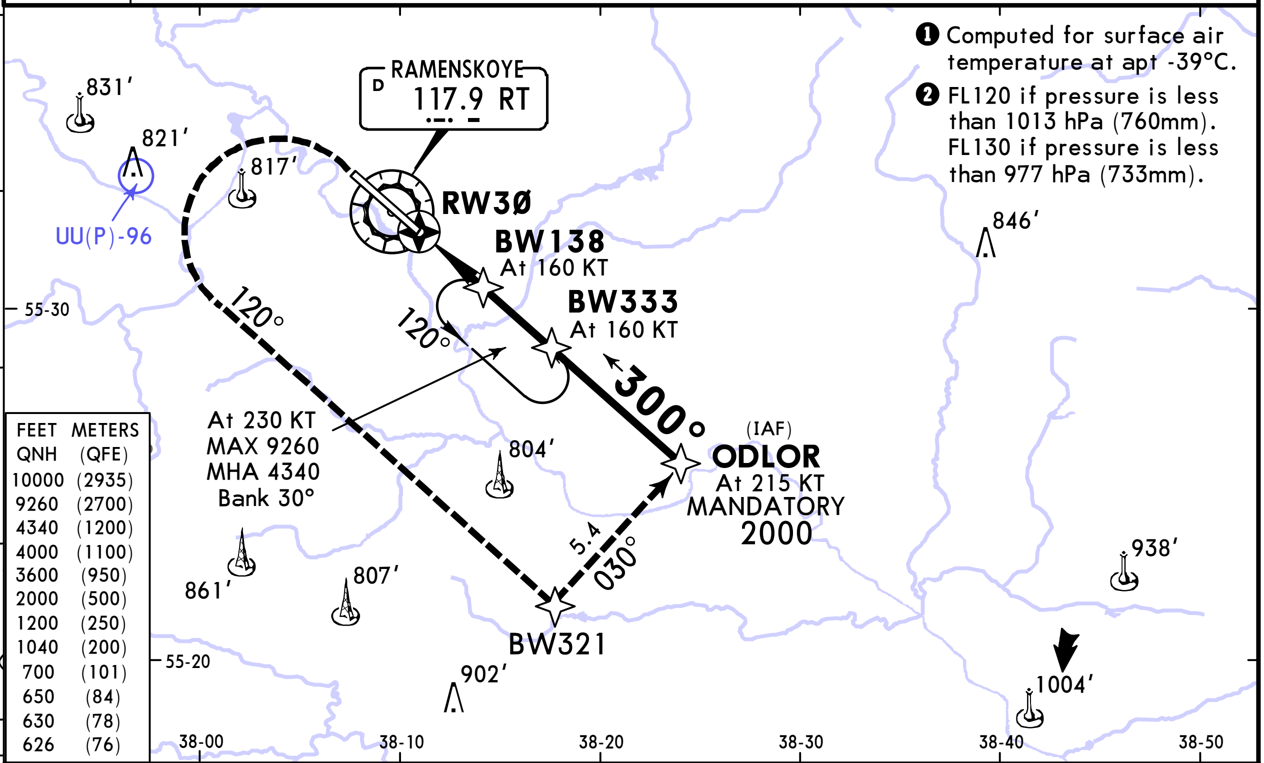
RNAV	Final Apch Crs 300°	BW333 MANDATORY 2000′(1624′)	LNAV/VNAV DA(H) Refer to Minimums	Apt Elev 404′ Rwy 376′



MISSED APCH: Climb STRAIGHT AHEAD to 1040′ or above (at 190 KT), then turn LEFT to BW321 (at 215 KT) climbing to 2000′, then proceed to ODLOR (at 215 KT) for instrument approach, or as directed.

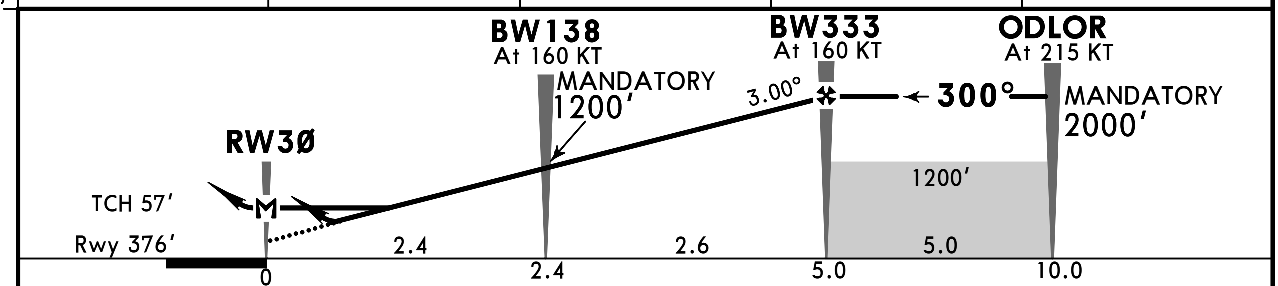
Alt Set: hPa (MM on req) | Rwy Elev: 14 hPa | Trans level: FL110 2 | Trans alt: 10000′

RNP apch. 1. GNSS required. 2. Baro-VNAV not authorized below -31°C.



FEET QNH	METERS (QFE)
10000	(2935)
9260	(2700)
4340	(1200)
4000	(1100)
3600	(950)
2000	(500)
1200	(250)
1040	(200)
700	(101)
650	(84)
630	(78)
626	(76)

DIST to BW138	0.5	1.1	1.6	2.2
ALT	1360′	1540′	1720′	1890′
DIST to RW30	0.5	1.1	1.6	2.2
ALT	610′	770′	950′	1120′



Gnd speed-Kts		70	90	100	120	140	160
Glide Path Angle	3.00°	372	478	531	637	743	849
MAP at RW30							

HIALS-II PAPI

MIN 1040′ | 190 KT

Std STRAIGHT-IN LANDING

LNAV/VNAV		LNAV CDFA	
DA(H) AB: 626′(250′) C: 630′(254′) D: 650′(274′)		1 DA/MDA(H) 700′(324′)	
	ALS out		ALS out
A B C D: R750m	R1300m	R800m	R1500m

PANS OPS

1 VNAV DA(H) in lieu of MDA(H) depends on operator policy.

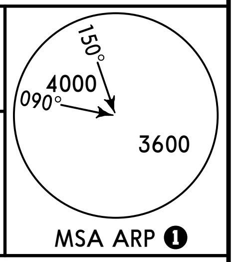
CHANGES: Note.

BRIEFING STRIP ™

*GORDY Approach/Radar	*GORDY Tower	*Ground
125.25	125.25	131.0

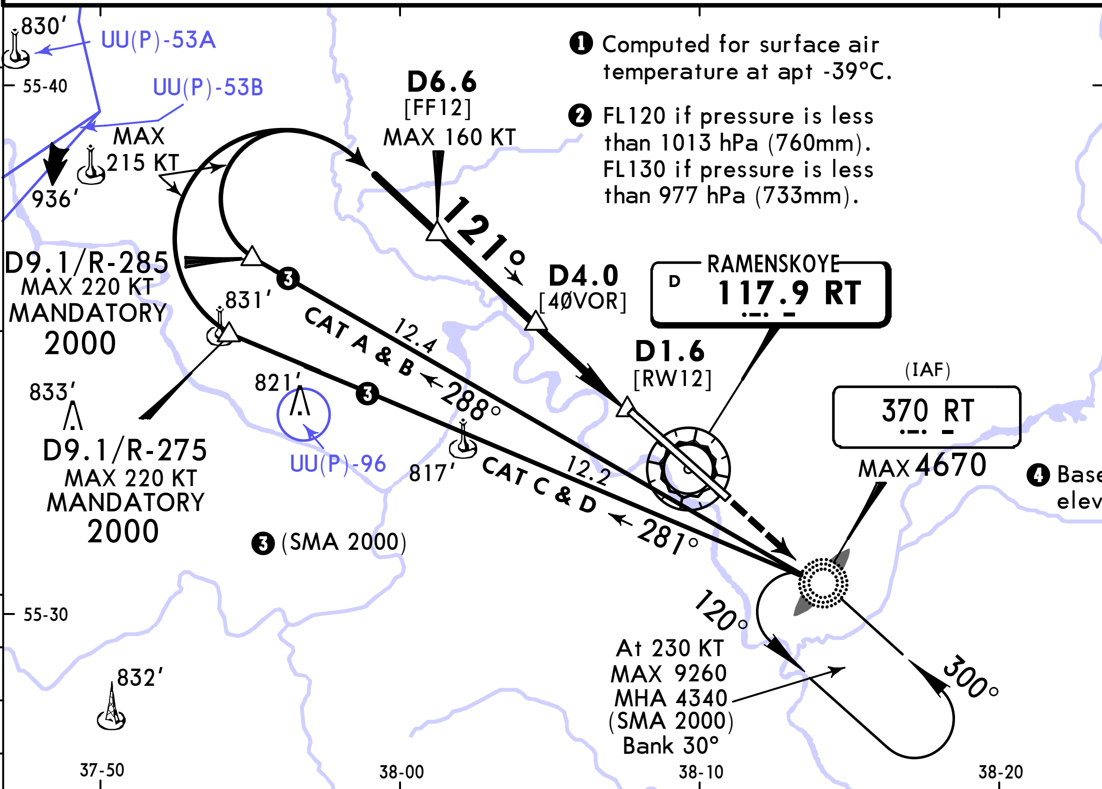
VOR RT	Final Apch Crs	D6.6 MANDATORY	DA/MDA(H)	Apt Elev 404' Rwy 401'
117.9	121°	2000' (1599')	850' (449')	

MISSED APCH: Climb STRAIGHT AHEAD to 1060' to NDB, then proceed to holding area, climbing to 6300' (at 230 KT), or as directed.



Alt Set: hPa (MM on req) Rwy Elev: 15 hPa Trans level: FL110 ❷ Trans alt: 10000'

1. DME required. 2. Final approach track is offset by 1°.



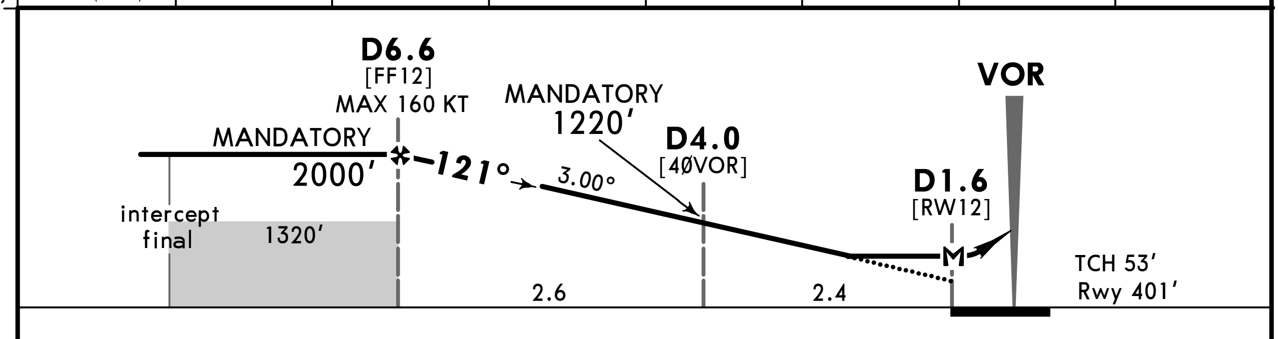
❶ Computed for surface air temperature at apt -39°C.

❷ FL120 if pressure is less than 1013 hPa (760mm). FL130 if pressure is less than 977 hPa (733mm).

FEET QNH	METERS (QFE)
10000	(2935)
9260	(2700)
6300	(1800)
4670	(1300)
4340	(1200)
4000	(1100)
3600	(950)
2000	(500)
1320	(280)
1220	(250)
1160	(232) ❹
1060	(200)
1040	(195) ❹
980	(177) ❹
850	(137)

❹ Based on airport elevation.

RT DME	6.6	5.9	5.3	4.8	4.3	3.7	3.2
ALT(HAT)	2000'	1830'	1670'	1490'	1320'	1140'	980'



Gnd speed-Kts	70	90	100	120	140	160
Descent Angle 3.00°	372	478	531	637	743	849
MAP at D1.6						

PAPI-L

1060' ↑

Std

	STRAIGHT-IN LANDING	Max Kts	CIRCLE-TO-LAND MDA(H)	
	CDFA **1** DA/MDA(H) 850' (449')			
A	R1500m	100	980' (576')	V1500m
B	R1500m	135	1040' (636')	V1600m
C	R2100m	180	1160' (756')	V2400m
D	R2100m	205	1160' (756')	V3600m

1 VNAV DA(H) in lieu of MDA(H) depends on operator policy.

PANS OPS

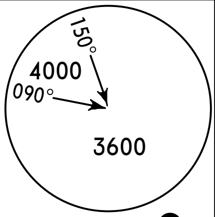
CHANGES: Communications.

BRIEFING STRIP ™

*GORDY Approach/Radar	*GORDY Tower	*Ground
125.25	125.25	131.0

VOR RT	Final Apch Crs	D5.9 MANDATORY	DA/MDA(H)	Apt Elev 404′ Rwy 376′
117.9	300°	2000′(1624′)	690′(314′)	

MISSED APCH: Climb STRAIGHT AHEAD to 1040′ (MAX 165 KT), then turn LEFT (CAT C & D: MAX 215 KT) to NDB climbing to 2000′. Then proceed to holding area at NDB climbing to 6280′ (at 230 KT) or as directed.

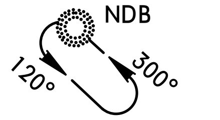


MSA ARP ❶

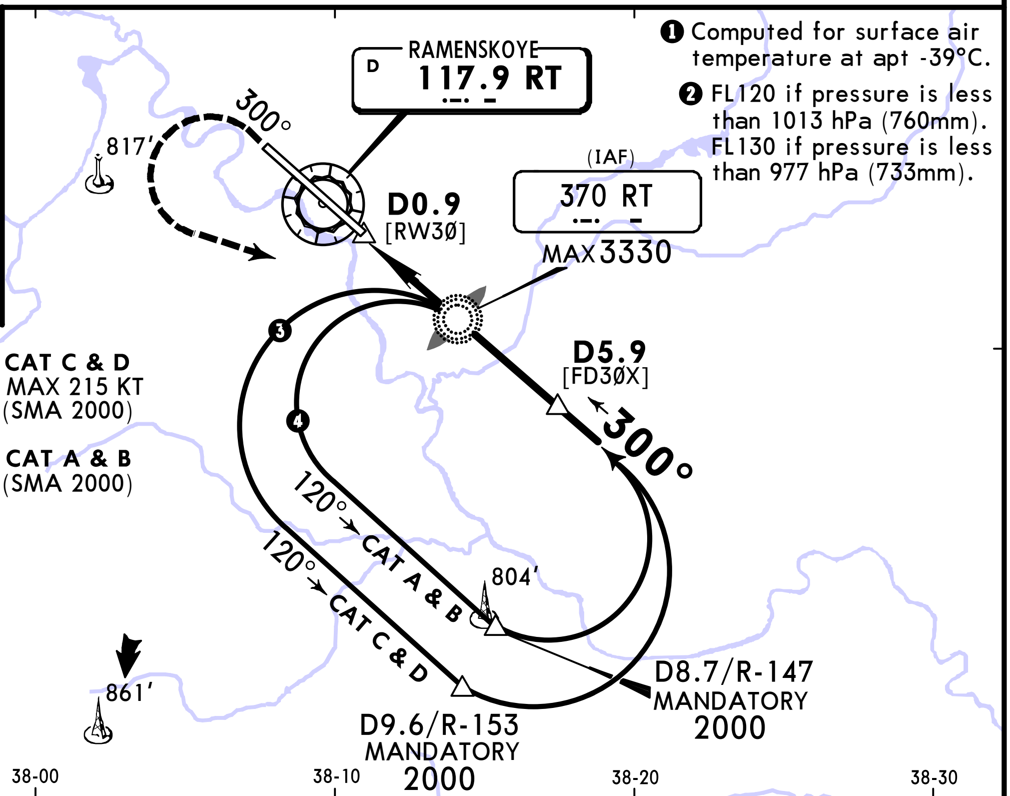
Alt Set: hPa (MM on req) Rwy Elev: 14 hPa Trans level: FL110 ❷ Trans alt: 10000′

DME required.

MISSED APCH HOLDING



At 230 KT
MAX 9260
MHA 4340
(SMA 2000)
Bank 30°



❶ Computed for surface air temperature at apt -39°C.

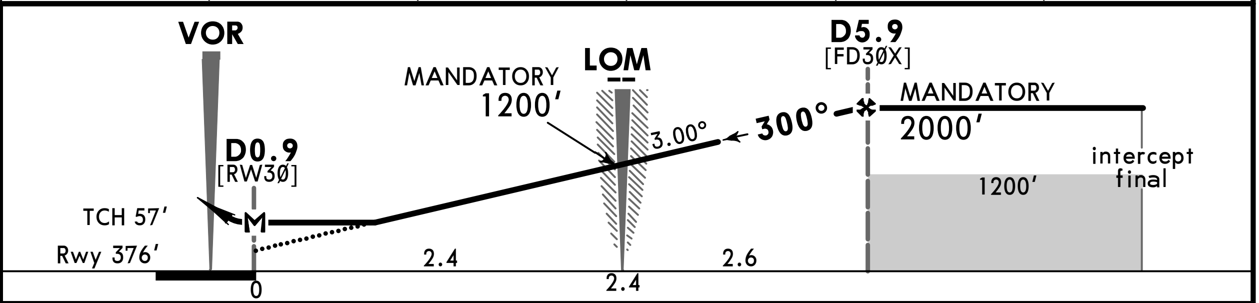
❷ FL120 if pressure is less than 1013 hPa (760mm). FL130 if pressure is less than 977 hPa (733mm).

❸ CAT C & D
MAX 215 KT
(SMA 2000)

❹ CAT A & B
(SMA 2000)

FEET	METERS
QNH	(QFE)
10000	(2935)
9260	(2700)
6280	(1800)
4340	(1200)
4000	(1100)
3600	(950)
3330	(900)
2000	(500)
1200	(250)
1040	(200)
690	(98)

RT DME	2.2	3.2	4.3	5.4	5.9
ALT(HAT)	800′	1160′	1500′	1850′	2000′



Gnd speed-Kts	70	90	100	120	140	160
Descent Angle 3.00°	372	478	531	637	743	849
MAP at D0.9						

HIALS-II
PAPI

1040′ ↑ MAX 165 KT

Std

STRAIGHT-IN LANDING

CDFA

1 DA/MDA(H) 690′(314′)

		ALS out
A	R750m	R1400m
B		
C		
D		

PANS OPS

1 VNAV DA(H) in lieu of MDA(H) depends on operator policy.

CHANGES: Communications.

UUBW/ZIA
RAMENSKOYE

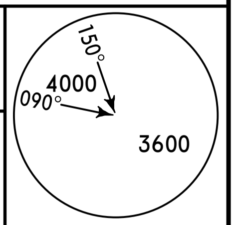
JEPPESEN
29 DEC 23 16-1

RAMENSKOYE, RUSSIA
NDB Y Rwy 12

*GORDY Approach/Radar	*GORDY Tower	*Ground
125.25	125.25	131.0

BRIEFING STRIP ™

NDB	Final Apch Crs	D6.6 MANDATORY	DA/MDA(H)	
D **1185**	**120°**	**2000'** (1599')	**880'** (479')	Apt Elev 404' Rwy 401'

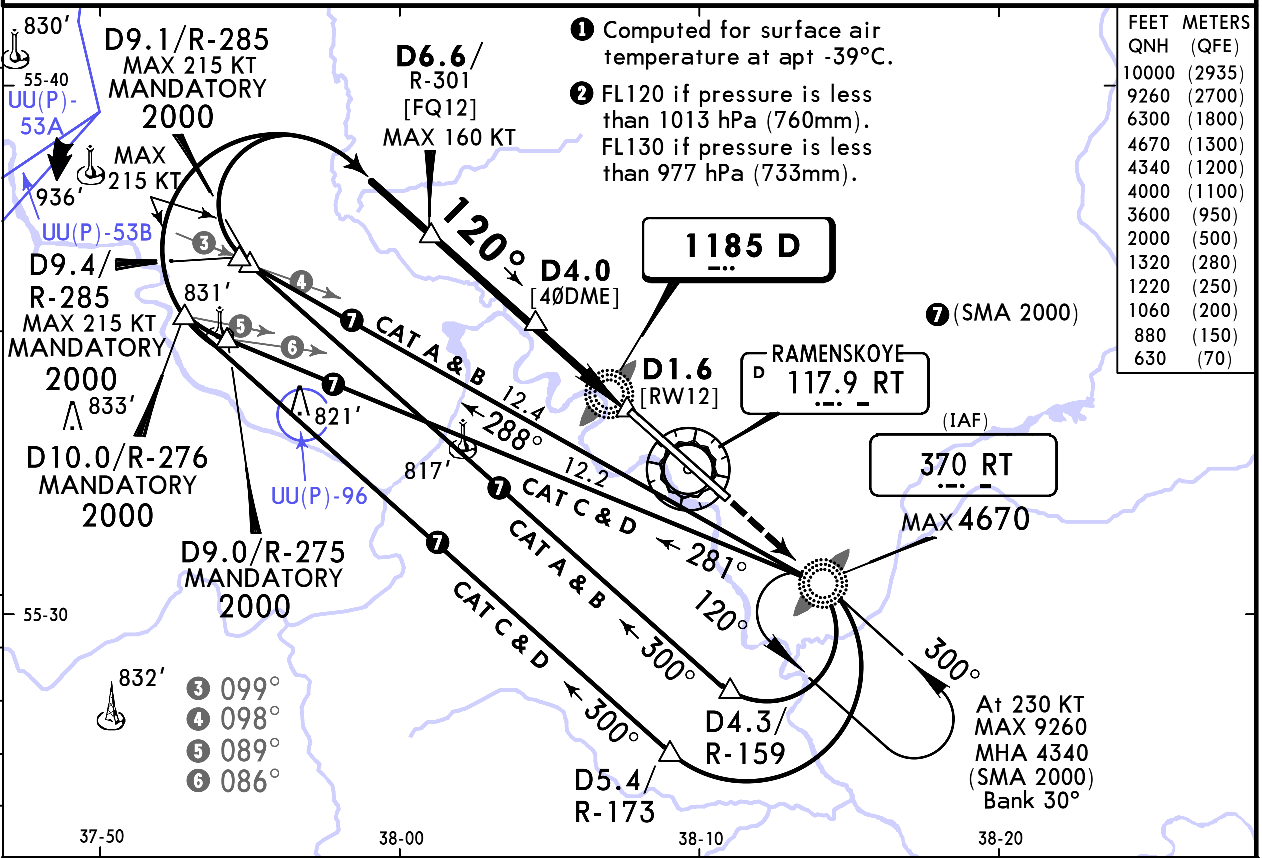


MSA ARP ❶

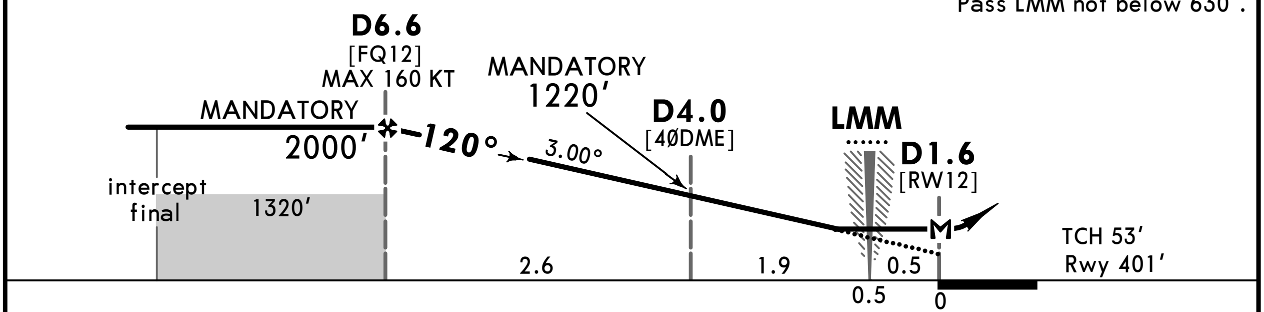
MISSED APCH: Climb STRAIGHT AHEAD to 1060' to RT NDB, then proceed to holding area RT NDB, climbing to 6300' (at 230 KT), or as directed.

Alt Set: hPa (MM on req)	Rwy Elev: 15 hPa	Trans level: FL110 ❷	Trans alt: 10000'

DME required.



Pass LMM not below 630'.



Gnd speed-Kts	70	90	100	120	140	160
Descent Angle 3.00°	372	478	531	637	743	849
MAP at D1.6						

PAPI-L

1060'

Std

STRAIGHT-IN LANDING

CDFA
❶ DA/MDA(H) **880'**(479')

A	R1500m
B	
C	R2200m
D	

PANS OPS

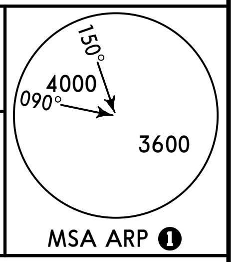
❶ VNAV DA(H) in lieu of MDA(H) depends on operator policy.

CHANGES: Communications.

*GORDY Approach/Radar	*GORDY Tower	*Ground
125.25	125.25	131.0

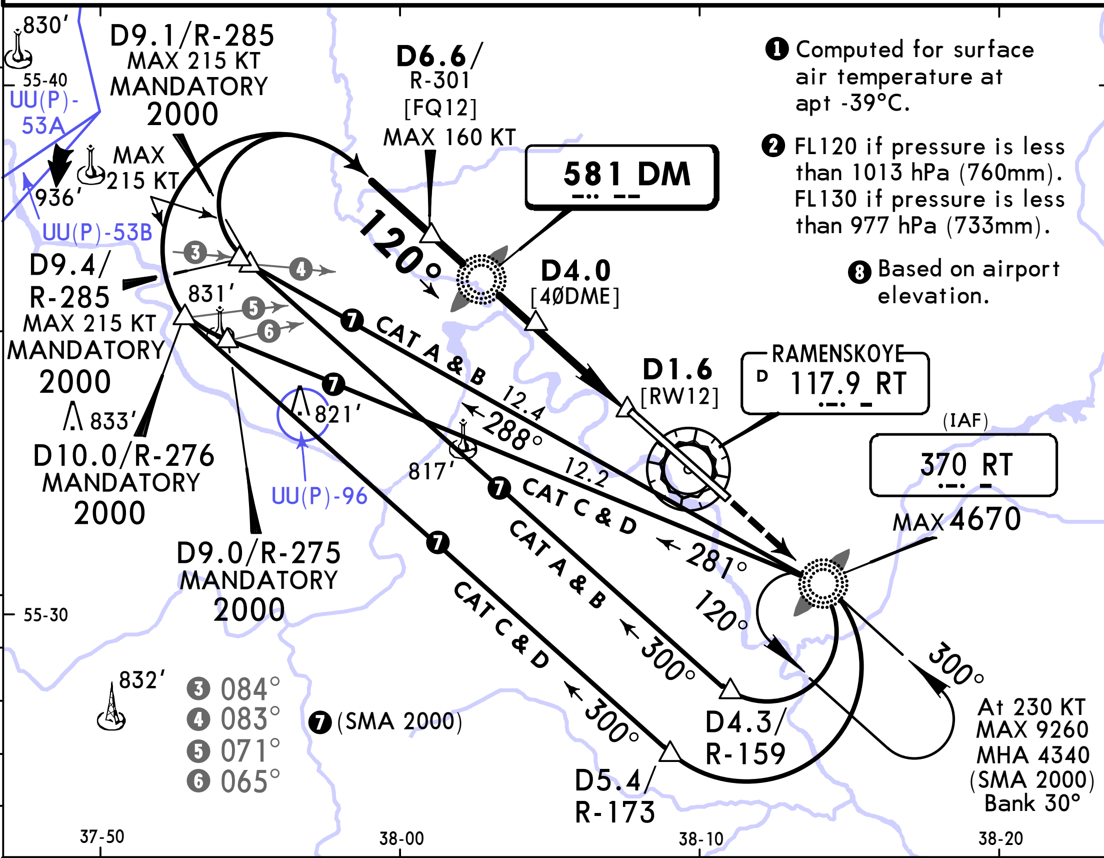
NDB DM	Final Apch Crs	D6.6 MANDATORY	DA/MDA(H)	Apt Elev 404'
581	120°	2000' (1599')	860' (459')	Rwy 401'

MISSED APCH: Climb STRAIGHT AHEAD to 1060' to RT NDB, then proceed to holding area RT NDB, climbing to 6300' (at 230 KT), or as directed.

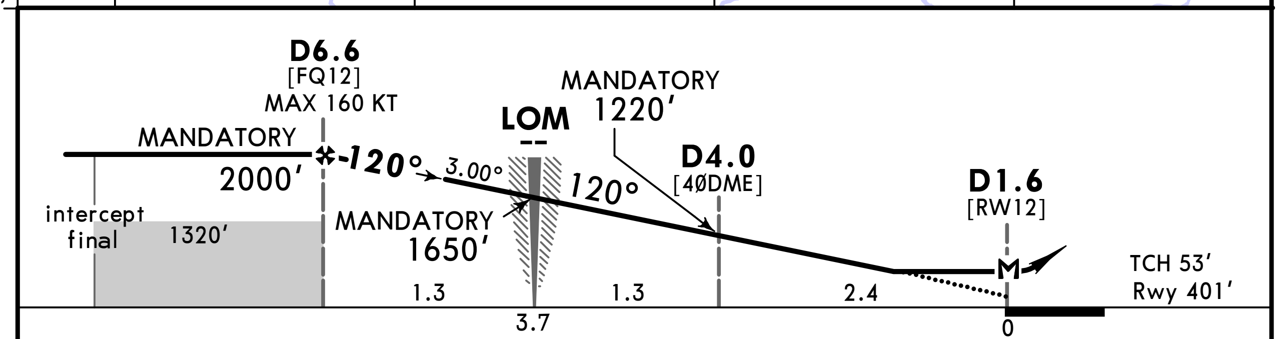


Alt Set: hPa (MM on req) | Rwy Elev: 15 hPa | Trans level: FL110 ❷ | Trans alt: 10000'

DME required.



FEET QNH	METERS (QFE)
10000	(2935)
9260	(2700)
6300	(1800)
4670	(1300)
4340	(1200)
4000	(1100)
3600	(950)
2000	(500)
1650	(380)
1320	(280)
1220	(250)
1160	(232) ❽
1060	(200)
1040	(195) ❽
980	(177) ❽
860	(139)



Gnd speed-Kts		70	90	100	120	140	160
Descent Angle	3.00°	372	478	531	637	743	849
MAP at D1.6							

PAPI-L | 1060' ↑

Std

	STRAIGHT-IN LANDING
	CDFA ❶ DA/MDA(H) **860'** (459')
A	R1500m
B	
C	R2100m
D	

CIRCLE-TO-LAND

Max Kts	MDA(H)	
100	980' (576')	V1500m
135	1040' (636')	V1600m
180	1160' (756')	V2400m
205	1160' (756')	V3600m

❶ VNAV DA(H) in lieu of MDA(H) depends on operator policy.

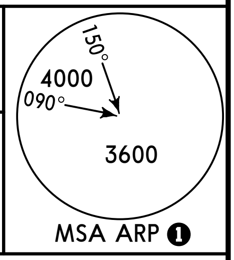
BRIEFING STRIP ™
PANS OPS

CHANGES: Communications.

BRIEFING STRIP ™

*GORDY Approach/Radar	*GORDY Tower	*Ground
125.25	125.25	131.0

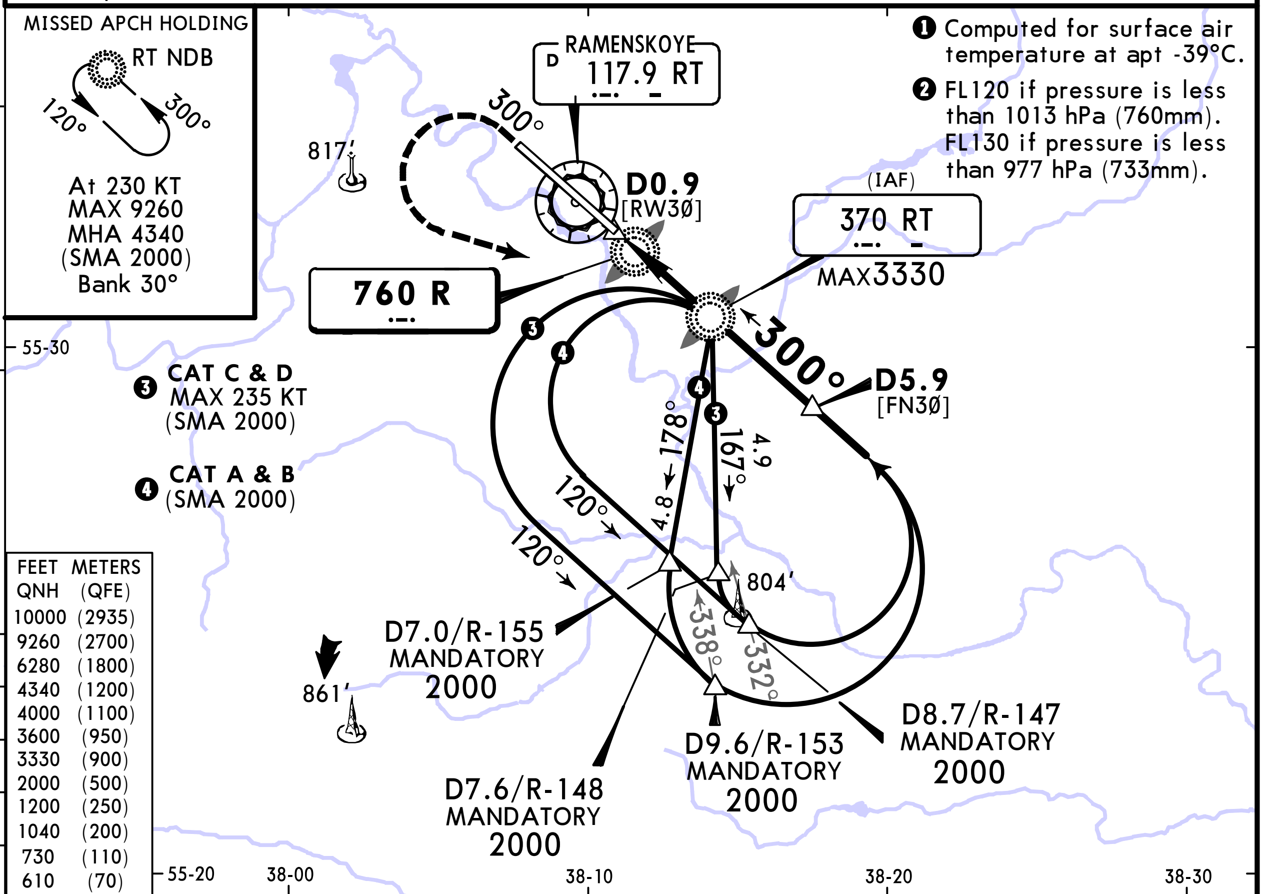
NDB	Final Apch Crs	D5.9 MANDATORY	DA/MDA(H)	Apt Elev 404'
R 760	300°	2000' (1624')	730' (354')	Rwy 376'

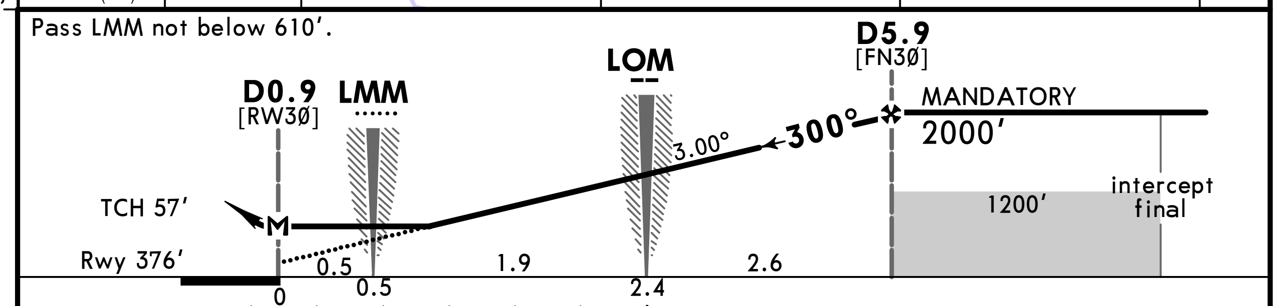


MISSED APCH: Climb STRAIGHT AHEAD to 1040' (MAX 165 KT), then turn LEFT (CAT C & D: MAX 235 KT) to RT NDB climbing to 2000'. Then proceed to holding area at RT NDB climbing to 6280' (at 230 KT), or as directed.

Alt Set: hPa (MM on req) | Rwy Elev: 14 hPa | Trans level: FL110 2 | Trans alt: 10000'

DME required.





Gnd speed-Kts	70	90	100	120	140	160
Descent Angle 3.00°	372	478	531	637	743	849
MAP at D0.9						

HIALS-II
PAPI

1040' ↑ | MAX 165 KT

Std

STRAIGHT-IN LANDING
CDFA
1 DA/MDA(H) 730' (354')

		ALS out
A	R900m	R1500m
B	R900m	R1500m
C	R900m	R1600m
D	R900m	R1600m

1 VNAV DA(H) in lieu of MDA(H) depends on operator policy.

PANS OPS

CHANGES: Communications.

UUBW/ZIA RAMENSKOYE

JEPPESEN

29 DEC 23 (16-4)

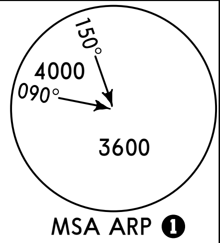
RAMENSKOYE, RUSSIA
NDB X Rwy 30

BRIEFING STRIP ™

*GORDY Approach/Radar	*GORDY Tower	*Ground
125.25	125.25	131.0

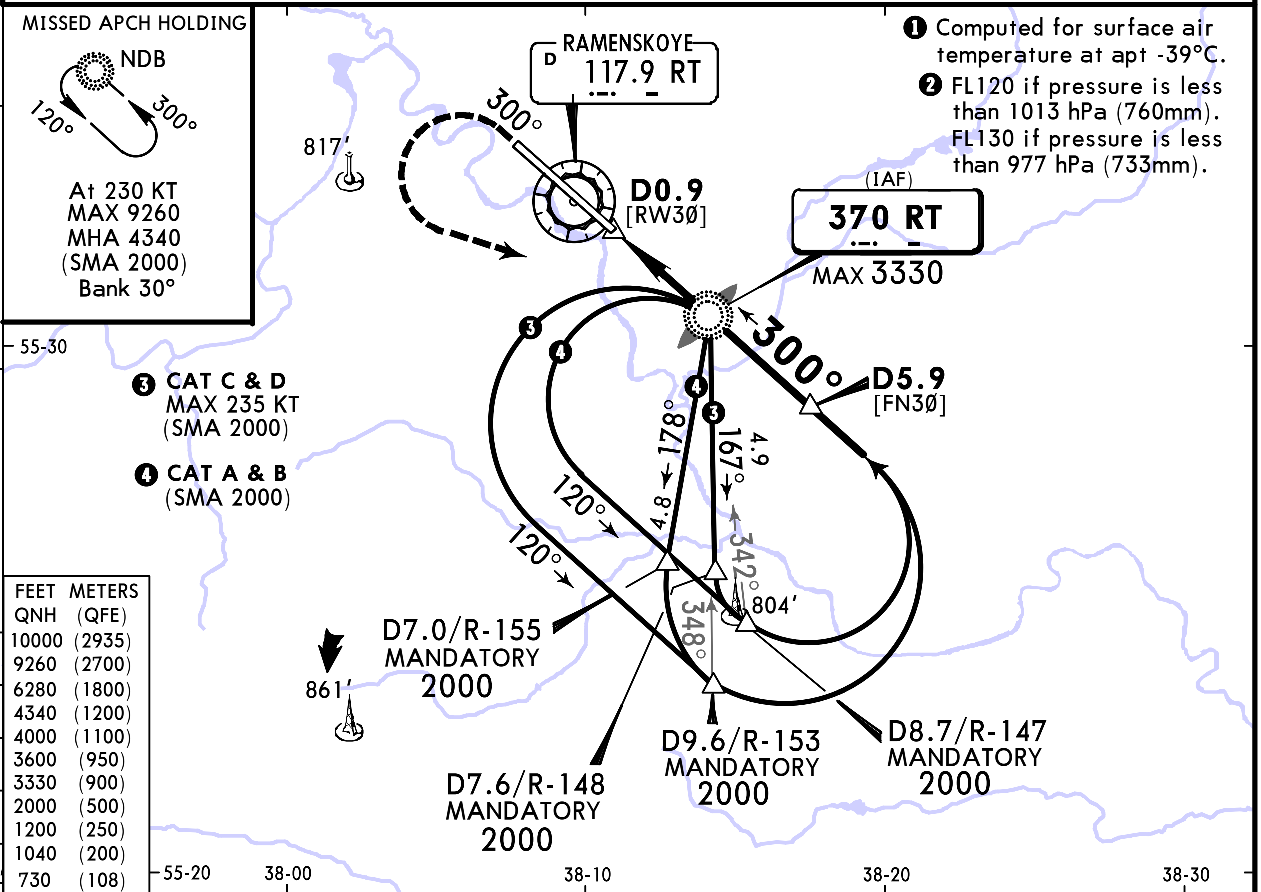
NDB RT	Final Apch Crs	D5.9 MANDATORY	DA/MDA(H)	Apt Elev 404′
370	300°	2000′(1624′)	730′(354′)	Rwy 376′

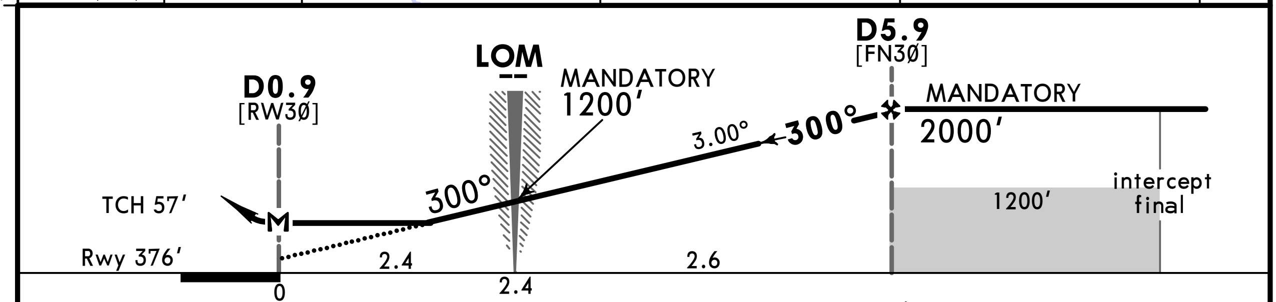
MISSED APCH: Climb STRAIGHT AHEAD to 1040′ (MAX 165 KT), then turn LEFT (CAT C & D: MAX 235 KT) to NDB climbing to 2000′. Then proceed to holding area at NDB climbing to 6280′ (at 230 KT), or as directed.



Alt Set: hPa (MM on req) | Rwy Elev: 14 hPa | Trans level: FL110 ❷ | Trans alt: 10000′

DME required.





Gnd speed-Kts	70	90	100	120	140	160
Descent Angle 3.00°	372	478	531	637	743	849
MAP at D0.9						

HIALS-II PAPI | 1040′ ↑ | MAX 165 KT

Std

STRAIGHT-IN LANDING
CDFA
❶ DA/MDA(H) 730′(354′)

		ALS out
A	R900m	R1500m
B		
C		R1600m
D		

❶ VNAV DA(H) in lieu of MDA(H) depends on operator policy.

PANS OPS

CHANGES: Communications.